The Social Multiplier of Pension Reform*

Emre Oral ⓡ Simon Rabaté ⓡ Arthur Seibold†

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Abstract

We study how social forces in families, neighborhoods and workplaces shape retirement behavior. To estimate causal retirement spillovers between individuals, we exploit a pension reform in the Netherlands that creates exogenous variation in peers' retirement ages, and we use administrative data on the full Dutch population. We find large spillovers in couples, primarily due to women reacting to their husband's retirement choices. Average spillovers among siblings, neighbors, and coworkers are modest; however, consistent with homophily in social interactions, sizable effects arise between similar individuals in these groups. Additional evidence suggests both leisure complementarities and the transmission of social norms as mechanisms behind retirement spillovers. Our findings imply that pension reforms have a large social multiplier, amplifying their overall impact on retirement behavior by at least 42%.

Keywords: retirement, pension reform, social networks, spillover, peer effects.
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†Author names appear in certified random order generated by the AEA author ordering tool; the confirmation code is hnbIXl2F138h. Oral: University of Mannheim, emre.oral@uni-mannheim.de. Rabaté: Institut National d'Études Démographiques (INED) and Centraal Planbureau (CPB), simon.rabate@ined.fr. Seibold: LMU Munich and CEPR, arthur.seibold@econ.lmu.de.

1 Introduction

Population aging poses a major challenge to the fiscal sustainability of public pension systems across the developed world. In OECD countries, the old-age dependency ratio is projected to more than double over the next 50 years (OECD, 2025). In response to these demographic pressures, many governments are enacting pension reforms aimed at encouraging later retirement. Understanding how retirement behavior responds to such reforms is crucial for assessing their fiscal impact and their welfare consequences.

In this paper, we investigate the role of social forces for retirement behavior. The idea that social context matters for retirement decisions has long been prevalent in social sciences.[1] Conceptually, there are several reasons why one may expect workers' retirement behavior to be influenced by others around them. First, there could be leisure complementarities, such that individuals wish to retire together with their peers. Second, retirement is a complex, one-off decision where individuals may learn from others' choices. Third, social norms regarding "normal" retirement ages might play a role. These types of spillovers carry an important policy implication: Pension reforms can have a *social multiplier*, altering their impact on overall retirement behavior. Yet, so far there is little systematic evidence on social interactions in retirement, and what these imply for the effectiveness of pension reforms.[2]

We aim at filling this gap in the literature and study to what extent retirement behavior is influenced by social networks in the context of a Dutch pension reform. We uncover significant retirement spillovers within families, neighborhoods, and workplaces. Additional evidence suggests that spillovers are driven both by leisure complementarities and by the transmission of social norms. Our findings imply a large social multiplier of pension reforms. Taken together, estimated spillover effects exacerbate the impact of a reform on the average retirement age by at least 42%, relative to its direct effect on individual behavior.

Quantifying causal retirement spillovers between individuals is challenging for two reasons. First, there are well-known econometric issues in identifying such peer effects, including correlated unobservables, endogenous group membership, and reflection problems (Manski, 1993; Sacerdote, 2014). Second, data requirements are very demanding: one needs information not only on retirement behavior, but also on the social linkages between individuals.

Our setting and data allow us to overcome these challenges. We exploit a pension reform in the Netherlands that generates large-scale, exogenous variation in peers' retirement behavior. The 2006 reform made early retirement much less financially attractive. The reform was implemented with a sharp cohort cutoff, affecting only workers born in January 1950 or later. As a result, average retirement ages increase discontinuously by around four months among directly affected individuals. While many other pension reforms tend to be implemented more slowly and gradually, this

[1] For instance, classic sociological work by Atchley (1982) examines the social processes shaping retirement behavior over the course of modern history and ultimately characterizes retirement as a *social institution*.

[2] An exception is given by a number of studies examining interactions in the retirement behavior of couples (see below).

sharp, immediate variation makes the Dutch setting ideally suited for our purposes.

We use high-quality administrative data provided by Statistics Netherlands (CBS), covering the universe of Dutch residents. Crucially, this data allows us to observe individuals' social linkages within families, neighborhoods, and workplaces. We merge various administrative registers at the individual level, including civil registration data, employer-employee matched labor market data, and income tax data. Taken together, the dataset we assemble contains detailed information on the retirement behavior of all workers linked to their spouses, siblings, neighbors and coworkers.

Our analysis relies on two complementary empirical strategies, both of which exploit exogenous variation in peers' retirement behavior around the cohort cutoff of the 2006 reform. The first strategy is a two-stage regression discontinuity (RD) design. In the first stage, we estimate the direct effect of the reform on peers' retirement age. In the second stage, we estimate the causal spillover effect of peers' reform exposure on the retirement behavior of other individuals. As a second empirical strategy, we use an instrumental variable (IV) approach exploiting variation in the fraction of an individual's peers treated by the reform. Although this strategy relies on a somewhat stronger identification assumption than the RD design, it offers the advantage of utilizing the full variation in the reform exposure of peer groups. This feature is particularly useful for large groups, such as neighbor and coworker networks, where the IV approach provides substantially greater statistical power.

We begin by documenting retirement spillovers in couples. We find strong spillover effects of spouses' reform exposure. On average, individuals increase their retirement age by 6.3 months for each year their spouse delays retirement due to the reform. These effects are primarily concentrated among women, who respond by 11.3 months per year of later retirement of their husband. Conversely, men exhibit no significant reaction to their wife's reform exposure. Further analyses reveal that neither age differences within couples nor relative earnings can explain this gender asymmetry in spillovers. While couples' retirement has been studied in prior work, this analysis provides a useful benchmark for the rest of our analysis, and spousal influences form an important part of our multiplier calculation.

We then provide a comprehensive analysis of retirement spillovers across the social network beyond couples. First, we examine siblings, the closest non-spousal family members within the same generation. On average, the impact of a sibling's reform exposure on individuals' retirement behavior is small and insignificant. Large spillovers emerge, however, among similar sibling pairs who share socioeconomic characteristics such as gender, pre-retirement earnings, city of residence, and sector of work. For these sub-samples, we find large spillover effects of up to 8.4 months.

Next, we investigate retirement spillovers among neighbors. Using variation in the reform exposure of all neighbors, we find modest but precisely estimated spillovers of 0.7 months per year of peers' later retirement. Substantially larger spillovers of 2.2 months occur when an individual's most similar neighbor sharing key socioeconomic characteristics retires later. We also show that spillovers between neighbors increase in local population density, suggesting that the frequency of social interactions matters for the magnitude of peer influence.

Finally, we turn to coworkers, where our findings closely mirror those for the other non-spousal peers. On average, spillover effects among all coworkers are modest at 1.3 months per year of peers' later retirement. Among the most similar coworkers, we find that individual retirement ages react by 3.5 months to a one-year retirement delay of the peer. Furthermore, we find that spillovers are particularly strong in small firms, again indicating that peer influence increases in the intensity of social interactions.

This comprehensive analysis of spillovers enables us to systematically quantify the influence of multiple dimensions of the social network on retirement behavior. Our results demonstrate that, in addition to spouses, many types of non-spousal peers exert significant spillover effects on retirement. Moreover, because we estimate these spillovers within a common setting, we can gauge the relative impacts of different social relations on retirement. As might be expected, the influence of the average non-spousal peer is modest relative to the benchmark of couples. Among these peers, our main estimates suggest that coworkers exert a larger average influence than neighbors and siblings. Consistently across all non-spousal peer groups, we find that very similar siblings, neighbors or coworkers are highly influential, with effects reaching more than half of the average spousal effect. This similarity gradient points to *homophily* in social networks, that is, the tendency of individuals to interact and form social ties with others who are similar to them (McPherson et al., 2001).

These estimates of total spillover effects are relevant from a policy perspective, as they govern the overall behavioral and fiscal externalities of pension reforms. Nevertheless, we also explore the mechanisms underlying retirement peer effects. First, we argue that information spillovers cannot explain our results. In principle, reform-induced changes in peers' behavior could convey information about the post-reform retirement rules. However, we find that spillover effects are not driven by individuals for whom this information would be valuable. Instead, we show that leisure complementarities matter, especially in couples. Spillover effects are particularly large among pairs who coordinate their retirement timing. But since the fraction of joint retirements is limited, leisure complementarities cannot account for the majority of spillovers.

Moreover, we provide evidence in favor of the transmission of social norms. In the Netherlands, there is a large concentration of retirements at the Normal Retirement Age (NRA), even though there is no financial incentive to retire at this age. Prior literature argues that this type of behavior can be explained by the widespread perception of the NRA as a "normal" time to retire (Behaghel and Blau, 2012; Seibold, 2021, 2024). We demonstrate that this specific, norm-driven behavior is propagated via social spillovers: individuals become more likely to retire exactly at the NRA if their spouse, neighbor or coworker is induced to retire at the NRA by the reform. We also show that behavioral spillovers occur mostly among individuals who face no own incentive to retire later, further supporting social norms as a mechanism underlying our main results.

Finally, we quantify the policy implications of our results. To calculate the social multiplier of pension reform, we aggregate the estimated spillover effects across all peer groups in our data. We find a large social multiplier of at least 1.42. In other words, when a pension reform increases the

average retirement age of a population by one year through its direct effect, social spillovers amplify its impact by an additional 0.42 years. Similarly, spillovers exacerbate the effect of the reform on the government budget by at least 34%. This indicates that standard individual-level analyses of pension reforms significantly under-estimate their behavioral and fiscal effects. Ultimately, our findings imply that the presence of a social multiplier can substantially alter the welfare impact of pension reforms, for which total fiscal externalities arising from behavioral responses are a key determinant (Kolsrud et al., 2024).

This paper contributes to a large and growing literature on retirement behavior and pension reforms. Much of this literature focuses on individual retirement decisions and how they respond to pension policies (see Giupponi and Seibold, 2024, for an overview). This includes responses to financial retirement incentives (Brown, 2013; Manoli and Weber, 2016; Duggan et al., 2023) and statutory retirement ages (Behaghel and Blau, 2012; Staubli and Zweimüller, 2013; Rabaté, 2019; Seibold, 2021; Gruber et al., 2022; Dolls and Krolage, 2023; Lalive et al., 2023; Deshpande et al., 2024). The impact of the Dutch 2006 pension reform on individual retirement is analyzed by Lindeboom and Montizaan (2020) and Rabaté et al. (2024).[3]

In contrast, much less is known about the role of social forces in shaping retirement behavior. An exception is given by retirement interactions in couples, which are studied by several papers (e.g. Atalay et al., 2019; Johnsen et al., 2022; Nagore Garcia and van Soest, 2021, 2022; Garcia-Miralles and Leganza, 2024).[4] So far, there is particularly little evidence on the influence of non-spousal peers on retirement decisions. To our knowledge, the only existing study is Brown and Laschever (2012) who investigate retirement peer effects in a sample of Los Angeles high-school teachers.[5]

We make three main contributions to this literature. First, we map retirement spillovers across the full social network observable in administrative data. In particular, we provide novel evidence of retirement spillovers between siblings and neighbors, and the first large-scale evidence of retirement spillovers in workplaces. Our results show that these non-spousal peers have economically and statistically significant impacts on retirement behavior. Moreover, we document the important role of homophily in shaping these peer effects, a feature that has received little attention in prior literature.

Second, our results shed new light on the mechanisms underlying retirement spillovers, in particular on social norms. While prior literature has theoretically considered the potential role of social norms for retirement (e.g. Axtell and Epstein, 1999), direct empirical evidence on their transmission has been scarce. We extend the recent literature on statutory retirement ages by providing the first evidence that their behavioral impact is mediated by social norms.

Third, building on our comprehensive analysis, we quantify the overall policy implications

[3]Moreover, De Grip et al. (2012) and Montizaan and Vendrik (2014) study the effect of this reform on mental health and job satisfaction.

[4]Couples' retirement behavior is also examined by Zweimüller et al. (1996), Lalive and Parotta (2017), Selin (2017), and Bloemen et al. (2019).

[5]A working paper by Chalmers et al. (2008) also investigates peer effects among public employees in Oregon.

of social interactions in retirement behavior. Our social multiplier calculations imply that social spillovers significantly amplify the behavioral and fiscal effects of pension reforms. This multiplier effect has not been taken into account in existing fiscal and welfare analyses of retirement policies.

More broadly, this paper contributes to the literature on peer effects in social insurance programs. Most closely related is work on social interactions in savings decisions in the context of pension plans (e.g. Duflo and Saez, 2002, 2003; Beshears et al., 2015). Prior work also studies social interactions in disability insurance (Rege et al., 2012; Dahl et al., 2014a; Dahl and Gielen, 2021), health insurance (Sorensen, 2006; Handel et al., 2020), unemployment insurance (Kroft, 2008; Markussen and Røed, 2015), social assistance (Bertrand et al., 2000; Åslund and Fredriksson, 2009), and parental leave benefits (Dahl et al., 2014b). Our main contribution to this wider literature is to analyze social spillovers in retirement decisions – one of the most consequential behavioral margins for social security programs – and to do so across multiple dimensions of the social network.

The remainder of this paper is organized as follows. Section 2 outlines context and data, Section 3 describes our empirical strategy, Section 4 presents the main results, Section 5 examines mechanisms behind spillover effects, Section 6 discusses policy implications, and finally, Section 7 concludes.

2 Context and Data

2.1 The Dutch Pension System

The Dutch pension system consists of three pillars. The first pillar is given by public pay-as-you-go pensions, which are financed by social insurance contributions. Public pension benefits depend only on how long a person has been resident in the Netherlands. Individuals accumulate two percent of the full benefit amount per year of residence, up to the full amount which is paid to those with at least 50 years of residence. Throughout our sample period, the full benefit amount is tied to the national monthly minimum wage ("social minimum").[6] Public pensions are automatically claimed at the Normal Retirement Age (NRA, or *AOW* age in Dutch). The NRA was set at 65 for decades, but has been increased to 67 in recent years. It is worth noting that many employment contracts end automatically at the NRA, such that firms and workers have to actively renew the contract in order to continue employment.

The second pillar is employer-based pension schemes. Participation in these schemes is mandatory for all employees. Most second-pillar plans are fully funded and organized at the level of sectors, except a few cases where firms have their own schemes. Contribution payments are made by workers and firms, and benefit amounts are directly linked to contributions. Second-pillar pensions can be claimed from an Early Retirement Age (ERA), with some actuarial adjustment of monthly benefit amounts. Specific benefit and retirement rules, including the ERA, can differ

[6] A retired single individual receives 70% of the monthly minimum wage, whereas couples jointly receive 100% of the monthly minimum wage. The monthly minimum is set by the government and updated twice per year. As of July 2023, the monthly minimum wage was EUR 1995.

across sectors. Employer-based pensions are an important component of old-age income in the Netherlands. The median individual receives a replacement rate from second-pillar pensions of 29% in addition to a median replacement rate of 39% from public pensions (Knoef et al., 2017).

The third pillar of the pension system is given by voluntary private pension plans. Contributions to individual retirement savings accounts are encouraged by the government via a tax credit. Partly due to the large and well-established first- and second-pillar schemes, private pension savings play a less important role, contributing about 5% of overall pension income on average (Lindeboom and Montizaan, 2020).

2.2 The 2006 Reform

In the early 2000s, there were growing concerns about the sustainability of the Dutch pension system. At the time, early retirement was the norm in the Netherlands: around 80% of workers retired at the age of 62 or younger (Lindeboom and Montizaan, 2020). Early retirement was largely facilitated by employer-based pensions, which provide the only source of pension income before the NRA for most workers. These concerns prompted a major reform to second-pillar pension rules in 2006. The main goal of the 2006 reform was to incentivize later retirement by making early retirement provisions less generous. The reform thus comprised a bundle of rule changes to employer-based pensions. This included a cut in the level of early retirement benefits and the introduction of actuarially fair benefit adjustment, strengthening marginal financial incentives for later retirement. Moreover, specific early retirement provisions, such as "bridge" payments before the NRA, were abolished. Finally, a minimum ERA of 60 was legislated. The reform was announced at relatively short notice in July 2005, only a few months before taking effect.

Importantly for our empirical strategy, the 2006 reform was implemented with a sharp cohort cutoff. Only workers born on January 1, 1950 or later were affected, whereas older workers born below the cutoff could still retire under the generous pre-reform early retirement rules. Taken together, the bundle of reform measures caused a large increase in average retirement ages among workers born above the January 1950 cutoff (see Section 4.1).

Other Reforms. A few other policy changes that occurred during our sample period are worth noting. First and largest, the NRA increased from 65 to 67, which was phased in between 2013 and 2024 (see Rabaté et al., 2024). The NRA gradually increased at a set of cutoff dates based on calendar years, none of which coincide with the January 1950 cohort cutoff of the 2006 reform. Second, a pension supplement for non-working spouses of public pension claimants ("partner allowance") was removed in 2015. The timing of this reform may affect some individuals in our data, but prior literature finds no major impact of this policy on individual retirement around the 1950 cutoff (Nagore Garcia and van Soest, 2021, 2022). While in place, however, the partner allowance may have created incentives for joint retirement of couples, which we discuss further in Section 5. Finally, smaller reforms during our sample period include the introduction of tax-free retirement savings accounts in 2006 and a subsequent reform restricting eligibility in 2015, as well

as the introduction of tax credits for working at old age in 2012 and their removal in 2018. Since these policy changes do not differentially affect individuals by cohort, they should not pose any concerns for our analysis of the 2006 reform.

2.3 Data

We use rich administrative data on the universe of Dutch residents provided by Statistics Netherlands (CBS). We construct our main dataset by merging a number of administrative registers at the individual level.[7] This includes civil registration data, personal income tax data, and employer-employee matched labor market data. We observe individual labor market states and incomes at the monthly level. Our main outcome of interest is a worker's retirement age, which we define as the age of last employment exit. For the analysis of retirement spillovers in families, we rely on information on household composition and family links from civil registers. Our analysis of workplaces is based on firm and sector of employment, which we observe in the labor market data. In order to define neighborhoods, we use residential locations from civil registers, which we observe at the level of fine-grained six-digit postcodes. Moreover, the administrative data contains some sociodemographic characteristics, including date of birth, gender, marital status and an indicator for foreign-born individuals.

Our analysis period is 2003 to 2023. We implement two main sample restrictions. First, we focus on individuals born between 1946 and 1954, such that our sample includes a number of cohorts around the reform cutoff for whom we can observe a retirement age range between 57 and 67. Second, we require that individuals are employed for at least one month at or after the age of 57. This ensures that individuals are attached to the labor market, such that we can interpret their last employment exit as a retirement event. We generally assign peer group membership before retirement. This is important in order to circumvent issues around potential retirement-related switches between peer groups, such as moving to another neighborhood around the time of retirement. Thus, we define neighborhoods based on place of residence at age 57 and couples as individuals who are married or in a registered civil partnership at age 57. Because of a change in firm identifier variables in the earlier years of data, we cannot define coworker networks for all individuals at age 57; we therefore construct them based on the firm where an individual is employed at age 59.

Panel A of Table 1 shows summary statistics of our full sample, which comprises 1,367,064 individuals. 39% of the sample are female, reflecting lower pre-retirement labor force participation among women than men in the cohorts we study. 76% of individuals are married or in a civil partnership and 86% are Dutch-born. Annual labor income at age 57 is EUR 34.9k on average. The average retirement age is 63.8 years. 16% of individuals retire exactly at the NRA, 60% retire before the NRA, and the remaining 24% retire after the NRA. 57% of the sample are born in January 1950 or later, such that they are directly exposed to the 2006 reform. Panels B to E of the table show additional information for the subsamples we use in the various parts of our analysis,

[7] See Appendix B for a detailed description of the administrative registers we use.

which are limited to individuals with at least one peer of the respective type within the sample. For instance, the Couples Sample consists of 479,418 individuals whose spouse also satisfies our sample inclusion criteria. In the Siblings Sample, the median individual has one sibling, whereas in the Neighbors and Coworkers Samples, the median individual has 57 and 95 potential peers, respectively. These differences in group sizes will matter for how we estimate retirement spillovers across different types of peers. In addition, we note that the size distribution of firms is more skewed than that of other groups, as can be seen from the large difference between the median and the average number of coworkers. Nevertheless, family, neighborhood and workplace peers are quite similar in terms of cohort composition and average retirement ages.

3 Empirical Strategy

Our primary objective is to estimate causal retirement spillovers in social networks. That is, we want to quantify the causal effect of peers' retirement behavior on the retirement behavior of an individual. It is well known that the empirical identification of such peer effects is challenging (Manski, 1993; Sacerdote, 2014). One set of challenges is given by "correlated effects": unobserved individual characteristics may be correlated with own retirement behavior and peer retirement behavior. This issue is especially likely to arise when individuals endogenously select into peer groups based on these characteristics. For instance, in our setting the selection of individuals into couples or workplaces may be related to unobserved retirement preferences. In addition, there can be an issue of reverse causality ("reflection problem") when individuals simultaneously influence each other's retirement behavior.

In order to address these crucial identification challenges, our approach is to exploit quasi-random variation in peers' retirement behavior generated by the 2006 pension reform. Specifically, we rely on two complementary empirical strategies: (1) a regression discontinuity design leveraging variation in the reform exposure of a single, specific peer, and (2) an instrumental-variable approach using variation in peers' reform exposure across larger groups.

3.1 Regression Discontinuity Design

Our first empirical strategy is a regression discontinuity (RD) design around the cohort cutoff of the reform. Building on Dahl et al. (2014b) who use a similar strategy to estimate peer effects in parental leave take-up, we implement the RD design in two stages. In the first stage, we estimate the direct effect of the reform on the retirement behavior of an individual j, who is a peer of individual i (i.e. a family member, neighbor, or coworker). In the second stage, we then estimate the spillover effect of j's reform exposure on i's retirement behavior.

The first-stage estimation equation is given by

$$R_j = \alpha_0 + \alpha_1 \mathbb{1}(d_j \geq 1950) + f_l(d_j)\mathbb{1}(d_j < 1950) + f_r(d_j)\mathbb{1}(d_j \geq 1950) + \varepsilon_j \tag{1}$$

where R_j is the retirement age of individual j, d_j is j's birth date, the running variable of the

RD design, $\mathbb{1}(d_j \geq 1950)$ is an indicator for being born in January 1950 or later, and f_l and f_r are local functions of birth date to the left and the right of the cutoff, respectively. Our baseline specifications parameterize f_l and f_r as local linear functions estimated separately on both sides of the cutoff, using an estimation bandwidth of 24 months on each side. Later on, we conduct extensive robustness checks for these and other specification choices.

The first-stage coefficient α_1 in equation (1) captures the *direct* effect of the 2006 reform on j's retirement behavior. In order to identify this causal effect, we require the standard RD identification assumptions. First, reform exposure should be as good as randomly assigned locally around the January 1950 cutoff. This requires in particular that the running variable cannot be manipulated. While manipulation of birth dates is of course unlikely, we nonetheless carry out the usual checks of the distribution of the running variable in Section 4. Second, no other policy change should occur around the January 1950 cutoff. This condition is satisfied by our institutional setting, as no other reforms are tied to the same cohort cutoff.

In the second stage of the RD design, we estimate spillover effects via the following equation:

$$R_i = \beta_0 + \beta_1 \mathbb{1}(d_j \geq 1950) + h_l(d_j)\mathbb{1}(d_j < 1950) + h_r(d_j)\mathbb{1}(d_j \geq 1950) + e_i \tag{2}$$

where R_i is the retirement age of individual i, and d_j is the birth date of another individual j, who is i's peer. As above, h_l and h_r are local functions of birth date below and above the January 1950 cutoff, respectively.

The coefficient β_1 captures the causal *spillover* effect of individual j's reform exposure (which induces variation in j's retirement age) on individual i's retirement age. Equation (2) thus yields an intention-to-treat (ITT) effect of j's reform exposure. In order to obtain a peer effect, i.e. an estimate of the causal impact of j's retirement age on i's retirement age, we can scale the reduced-form spillover effect β_1 by the first-stage estimate α_1.[8] Like the first stage, estimating spillover effects via equation (2) requires no manipulation of birth dates and no other policy changes coinciding with the January 1950 cutoff. In addition, identification relies on the absence of sharp sorting into peer groups by birth date locally around the cutoff. For instance, if there was a tendency to choose spouses born precisely in the same month, we may confound individuals' own reform exposure with that of their peer. We carefully examine these identification assumptions for each peer group in Section 4.

3.2 Instrumental Variable Approach

The RD strategy outlined above is well-suited for analyzing spillover effects of the retirement behavior of a single relevant peer. An important challenge that arises for some of the groups we study, however, is that individuals can have many potential peers, especially in neighborhoods and workplaces. In this case, implementing the RD design requires selecting one specific peer

[8] We follow Dahl et al. (2014b) and implement the second-stage as an ITT specification. This requires fewer assumptions than a fully-fledged two-stage least squares (2SLS) estimation. For instance, 2SLS would additionally require a monotonicity assumption that the reform does not cause any individuals to retire earlier.

for each individual. As an alternative strategy, we use an instrumental-variable (IV) approach leveraging variation in the reform exposure of multiple peers.[9]

Our starting point is a classic linear-in-means model of peer effects in the context of retirement decisions:

$$R_i = \gamma_0 + \gamma_1 \overline{R}_{-i} + X_i'\theta + \epsilon_i \tag{3}$$

where R_i is the retirement age of individual i, $\overline{R}_{-i}$ is the leave-out average retirement age of i's peers, and X_i is a vector of background characteristics. The coefficient of interest is γ_1, the causal effect of peers' retirement behavior on i's own retirement age.

The main idea underlying our IV approach is that the 2006 reform generates exogenous variation in the average retirement age of peer groups, depending on their local birth cohort composition around the January 1950 cutoff. Thus, we can utilize the reform cutoff to construct an IV for peers' average retirement age. Analogously to the RD design, we proceed in two stages. The first stage quantifies the direct effect of the reform on the average retirement behavior of individual i's peers, and the second stage estimates the spillover effect on i's own retirement behavior.

In the first stage, we estimate

$$\overline{R}_{-i} = \phi_0 + \phi_1 P(d_{-i} \geq 1950) + X_i'\zeta + \nu_i \tag{4}$$

where $P(d_{-i} \geq 1950)$ is the leave-out fraction of i's peers born above the January 1950 cutoff. The share of peers exposed to the reform thus serves as the IV, and its first-stage (direct) effect on peers' leave-out average retirement age is captured by coefficient ϕ_1.

In the second stage, we estimate the spillover effect on individual i's retirement behavior:

$$R_i = \delta_0 + \delta_1 P(d_{-i} \geq 1950) + X_i'\xi + u_i \tag{5}$$

Analogously to the RD equation (2), we estimate the second stage as an ITT specification directly regressing i's retirement age on peers' reform exposure. In order to scale spillover effects relative to the change in peers' average retirement age, we can divide the reduced-form effect δ_1 by the first-stage coefficient ϕ_1.

The IV strategy rests on two key identification assumptions. First, we need a sufficiently strong first stage, that is, the reform should induce significant variation in peers' average retirement behavior. As we show in the next section, estimating equation (4) indeed yields a highly significant first stage both among neighbors and coworkers. Second, we require peers' reform exposure to be as good as randomly assigned conditional on our controls. That is, there should be no selection into peer groups based on i's retirement preferences or other unobserved characteristics correlated with their retirement behavior. In particular, this rules out sorting into groups related to peers' cohort composition around the January 1950 cutoff.

Although the second identification assumption is somewhat stronger than its RD analogue, several arguments support its validity. First, we define reform exposure only within a local cohort

[9] Similar strategies that exploit variation in average reform exposure across large peer groups are used, for example, by Mora-Garcia and Rau (2023) and Hut (2024).

window around the reform cutoff, such that we use variation in the reform exposure of peer groups with similar overall age structure. Second, as explained in Section 2.3, we generally define group membership prior to retirement, limiting potential selection into peer groups related to retirement behavior. Third, in order to alleviate remaining concerns about sorting into peer groups based on age, we flexibly control for individuals' own birth date, which also accounts for their own reform exposure, and we include other pre-determined characteristics and fixed effects. Finally, we conduct placebo checks around artificial reform cutoffs in Sections 4.3 and 4.4. We find no significant correlation between individual retirement ages and the share of peers born above placebo cohort cutoffs, further supporting the validity of the IV approach.

4 Main Results

4.1 Direct Effect of the 2006 Reform

The 2006 reform serves as our source of exogenous variation in peers' retirement behavior. Panel (a) of Figure 1 documents the discontinuous impact of the reform on the retirement age of directly affected individuals. The figure corresponds to the first-stage RD specification from equation (1) run on the full sample. Consistent with prior work on the 2006 reform (Lindeboom and Montizaan, 2020; Rabaté et al., 2024), we observe a sharp jump in retirement ages occurring precisely at the January 1950 cutoff. Our RD estimate implies a highly significant effect of the reform on the average retirement age of directly affected individuals of 0.34 years (4.1 months). Section 4.5 presents an extensive set of robustness and specification checks for this first-stage RD result.[10]

In order to provide additional insight into the change in retirement patterns induced by the 2006 reform, Panel (b) of Figure 1 depicts the distribution of retirement ages among cohorts 1949 (the last year of birth below the reform cutoff) vs. 1950 (the first year above the cutoff). The figure shows that fewer individuals retire at ages 60 to 63 as a consequence of the reform. This is in line with our institutional setting, as the reform makes retiring at these early ages less attractive. Correspondingly, more individuals retire at later ages among treated cohorts. One especially noticeable pattern is the large spike in the density exactly at the NRA, which increases in magnitude among the post-reform cohorts. This growing bunching at the NRA seems to emerge as a side effect of the 2006 reform, which generally incentivizes later retirement but does not provide any incentive to retire precisely at the NRA. Similar retirement bunching at the NRA in other settings has been attributed to behavioral factors (Seibold, 2021). We further analyze this pattern when we discuss the role of social norms in Section 5.3.

Next, we turn to our main goal, leveraging this variation to estimate retirement spillovers in social networks.

[10]Moreover, Appendix Figure A1 shows that alternative retirement definitions, such as allowing for some earnings after retirement, requiring a minimum period out of employment, and using old-age pension claims rather than employment exits, yield very similar retirement patterns around the January 1950 cutoff. First-stage RD estimates are also similar under these alternative definitions, if anything becoming slightly larger.

4.2 Retirement Spillovers in Families

We begin our analysis of retirement spillovers with families. We focus on spouses and siblings, the closest types of family members within the same generation. Peer groups defined by families are relatively small: couples always consist of two individuals, and the majority of individuals have at most one sibling within our sample. Thus, the RD design described in Section 3.1 is well-suited for studying spillovers in families. After presenting our main results, we carefully assess the validity of the identification assumptions of the RD design and examine its robustness for couples as well as for all other peer groups in Section 4.5.

4.2.1 Spillovers in Couples

Our analysis of couples focuses on pairs of individuals who are married or in a civil partnership, where both spouses satisfy our sample inclusion criteria.[11] Figure 2 illustrates retirement spillovers in couples. Each panel provides a graphical depiction of equation (2), plotting individual retirement ages by the birth date of their spouse, the running variable. The text box included in each panel displays two estimates, the reduced-form RD estimate and the implied spillover effect scaled relative to the corresponding first stage within each sample.

Panel (a) of the figure shows average spillovers in couples. There is a sharp jump in individuals' average retirement ages at their spouse's birth date cutoff in January 1950. The RD estimate corresponds to a highly significant spillover effect of 0.17 years on individuals' own retirement age when their spouse is exposed to the reform. Next, Panels (b) and (c) display results for women and men separately. We find that women's retirement behavior is strongly impacted by their husband's reform exposure, with a spillover effect of 0.33 years. Conversely, there is no significant spillover effect on men when their wife is exposed to the reform. The point estimate of 0.02 is very small and not significantly different from zero. Thus, retirement spillovers within couples are almost exclusively driven by women reacting to their husband's retirement behavior.

Columns (1) to (3) of Table 2 show these RD results in more detail. In Panel A, first-stage estimates in the couples samples are highly significant, with correspondingly large F-statistic values. Interestingly, the magnitudes of the first stage in the male and female subsamples are quite similar, suggesting little gender difference in the direct effect of the reform. Panel B displays reduced-form spillover effects, and Panel C shows spillover effects scaled relative to the corresponding first stage in each sample. The scaled average spillover effect corresponds to a retirement age increase of 0.52 years (6.3 months) for each year the spouse retires later as a result of the reform. For women, the scaled spillover effect is 0.94 years and for men, the scaled effect of 0.07 years remains insignificant. This clear difference in scaled effects, together with the fact that first-stage estimates are similar in both samples, confirms the pronounced gender asymmetry in retirement spillovers from Figure 2.

Appendix Figure A2 presents additional results for couples, suggesting that this asymmetry cannot be readily explained by differences in age or earnings. First, we split the sample by gender

[11] Throughout this paper, we use the term "spouses" to refer to couples who are either married or in a civil partnership.

and spouses' relative age. We find that women who are younger than their husbands react most strongly to their husband's retirement age, whereas older women exhibit small spillover effects. In contrast, men's retirement choices never respond significantly to their wife, even when men are the younger spouse, indicating that relative age per se cannot explain the observed gender asymmetry. Second, we estimate spillover effects by gender and relative earnings within the couple. Women always react strongly to their husband's retirement age, regardless of whether they are the primary or secondary earner in the couple. Again, men are never significantly affected by their wife. This suggests that relative earnings cannot account for the gender asymmetry either. Overall, these results leave factors we do not observe directly, such as gender norms, as a plausible explanation.

Our finding of large retirement spillovers in couples is in line with prior studies (e.g. Johnsen et al., 2022; Garcia-Miralles and Leganza, 2024). Beyond establishing spousal spillovers in our setting, the couples results play two key roles in our subsequent analysis. First, they provide a natural benchmark for assessing the magnitude of spillovers arising from other social relations, for which the literature provides little guidance.[12] Second, couples are an important component of our social multiplier calculation in Section 6.

4.2.2 Spillovers between Siblings

Next, we investigate retirement spillovers between siblings. In order to implement the RD design, we focus on individuals who have one sibling within our sample. We exclude twins and other multiple births from the analysis, since own reform exposure and siblings' reform exposure perfectly coincide for these pairs, which would violate our identification assumptions.

Figure 3 displays our main results for siblings. Panel (a) considers all sibling pairs and plots individual retirement ages by the running variable, the sibling's birth date. Estimated retirement spillovers among all sibling pairs are positive but insignificant. However, we find strong spillovers among siblings who are similar to each other. In order to measure similarity, we use the main socioeconomic variables observed in our administrative data, namely gender, income, place of residence, and sector of work. Panel (b) shows a specification focusing on siblings who have the same gender, live in the same city, work in the same 2-digit sector, and who have similar earnings.[13] Note that all characteristics are measured prior to retirement, at age 57. For this group of very similar siblings, the RD estimate of 0.51 years is large and significant. Panel (c) shows a wider range of estimates across subsamples of siblings who share different sets of characteristics. We find that spillover effects increase systematically with siblings' degree of similarity. Spillovers become large and significant for pairs sharing two or more characteristics, in particular when sector of work and earnings are included. At the same time, confidence intervals widen for more similar sibling pairs, as these specific subsamples are relatively smaller in size.[14]

[12] A feature that potentially distinguishes spouses from other peers is the shared household budget constraint. As we discuss in Section 5, we find that these financial linkages are not the main driver of spillovers in couples, however.

[13] We define similar earnings as having labor earnings in the same decile of the labor earnings distribution among all workers.

[14] Note that the pattern in Figure 3 cannot be explained by selection of similar siblings into similar environments. Each

Columns (4) and (5) of Table 2 show full results from our preferred RD specifications for siblings. Both among all siblings and among the most similar pairs, we find a highly significant first stage. The first-stage point estimate is particularly large among similar siblings, but it is also more noisily estimated due to the relatively small size of this subsample. The reduced-form RD estimate among similar siblings translates into a scaled retirement age spillover of 0.70 years (8.4 months) for each year the sibling retires later. For the average sibling pair, the scaled spillover effect is 0.10 years (1.2 months) and not significantly different from zero.

Our findings for siblings are consistent with the notion of *homophily* in social networks, that is, the tendency of individuals to form closer social ties with others who are similar to them. A large body of interdisciplinary evidence suggests that homophily plays an important role in shaping social interactions across many domains (McPherson et al., 2001). In the context of retirement, similarity could amplify social spillovers through both exposure and influence. Similar individuals tend to have closer relationships and to interact more frequently, making it more likely that they discuss and observe each other's retirement choices. At the same time, individuals may perceive similar peers as more relevant to their own circumstances and thus put more weight on interactions with them.

4.3 Retirement Spillovers in Neighborhoods

Neighborhoods are an important part of individuals' social networks, and there is evidence that neighbors influence a number of economic decisions and outcomes (e.g. Luttmer, 2005; Chetty et al., 2013; Chetty and Hendren, 2018; Barrios-Fernandez, 2022). For our analysis of retirement spillovers between neighbors, a key consideration is that neighborhoods constitute large peer groups. Within our sample, the median individual has 57 potential peers in neighborhoods defined by 5-digit postcodes. We follow two empirical strategies to address this challenge. First, we employ the IV approach outlined in Section 3.2, which exploits the full variation in the reform exposure of neighborhoods in order to estimate average spillovers between neighbors. Our second strategy is an RD design similar to our approach for families, using specific neighbors as peers.

Average Neighborhood Spillovers. Table 3 shows IV results for neighbors. Column (1) begins with the first-stage estimation based on equation (4). We find a large and highly significant impact of the fraction of neighbors exposed to the reform on the leave-out average retirement age in the neighborhood. The point estimate suggests that increasing the share of treated neighbors by one standard deviation leads to an increase in peers' average retirement age by 0.12 years. The very large F-statistic underscores the key advantage of the IV strategy: it provides a large degree of statistical power by leveraging the full variation in reform exposure across neighborhoods. Next, Column (2) shows retirement spillovers estimated according to the reduced-form equation (5). We

estimate in Panel (c) compares individuals whose sibling was born just before with those whose sibling was born just after the reform cutoff, within a sample of siblings of a given degree of similarity. Causal spillover effects are therefore identified as long as the assumptions discussed in Section 3.1 hold for the respective sample.

find that a one-standard deviation higher neighbors' reform exposure causes a small but precisely estimated 0.007 year increase in individual retirement ages. The lower panel of the table reports scaled spillover effects implied by these estimates. We find that individuals retire 0.06 years (0.7 months) later per one-year increase in neighborhood peers' average retirement age.

A key identification assumption behind the IV approach is that individuals do not sort into residential areas based on neighbors' local cohort composition around the January 1950 threshold. As we discussed in Section 3.2, this concern is somewhat alleviated by assigning neighborhood peers before retirement and by accounting for age-based selection through controls for year-by-month of birth. Nevertheless, a remaining concern may be that individuals select into slightly younger neighborhoods (with a higher share of peers born above the cutoff) in ways correlated with their unobserved retirement preferences. In order to address this possibility, Column (7) of Table 3 shows a placebo IV specification around an artificial cohort cutoff in January 1948.[15] Apart from shifting the birth date cutoff, all specification choices are the same as in the main estimation. The placebo estimation yields a small and insignificant coefficient. This suggests that individual retirement ages are not generally correlated with neighbors' cohort composition, lending support to our identification assumption.

Influence of Specific Neighbors. As our second empirical strategy, we implement the RD design using specific neighbors as peers. This approach is motivated by our findings from the previous section, where we found an important role for homophily in retirement interactions. Concretely, we match individuals to their most similar neighborhood peer using the main socioeconomic characteristics contained in our administrative data, including gender, earnings, 2-digit sector of work, and an indicator for being foreign-born. As before, we measure these variables prior to retirement at age 57. We then construct similar-peer samples focusing on unique matches between individuals and peers who share the maximum number of characteristics.

Figure 4 displays results from this RD strategy. Panel (a) shows our preferred specification using the full set of characteristics to match individuals with their most similar neighbor. In this sample, we find a positive and significant spillover effect of the neighbor's reform exposure of 0.07 years. Column (7) of Table 2 shows these RD results in more detail. There is a strong first-stage effect of the 2006 reform on neighborhood peers' retirement ages in the similar-neighbor sample. Scaled relative to the first stage, the spillover effect exerted by similar neighbors corresponds to 0.18 years (2.2 months) per year of later retirement of the peer. Thus, the influence of the most similar neighbor on individuals' retirement is about three times larger than the average spillover effect from Table 3.

[15] An important challenge in implementing placebo specifications in our setting is that the Normal Retirement Age was increased at a set of cutoff dates within our sample period, potentially affecting retirement ages around candidate placebo cutoffs (see Section 2.2 and Rabaté et al., 2024). Indeed, a one-month increase in the NRA coincides with the January 1948 cutoff. For this reason, the RD placebo checks in Section 4.5 instead use an alternative placebo cutoff in January 1952. While there was no NRA increase exactly in January 1952, we do not use this cutoff here because the NRA increased at an accelerated pace in its vicinity, which would confound the placebo IV specification.

Spillovers by Population Density. How much neighbors influence each other's behavior may also depend on how close together they live. While the number of residents per postcode area is relatively uniform, there is substantial variation in local population density that we can exploit to this end. Panel (c) of Figure 4 shows retirement spillovers across five categories of population density defined by CBS, which we use to estimate heterogeneous spillover effects via our IV strategy (see Appendix Table A1 for detailed regression results). We find that spillovers are largest in the most densely populated areas. In the top two density categories, scaled spillover effects are at least 0.10 years (1.2 months) per one-year increase in the neighborhood's average retirement age, whereas in lower-density areas (categories 3 to 5) point estimates reach at most 0.03 years and are not statistically significant. That spillovers increase in population density is consistent with social interactions driving our results, as physical proximity between neighbors increases the likelihood of such interactions.

Additional RD Results. We show two additional RD results on neighbors in the appendix. First, Appendix Figure A3 includes an RD specification where peers are selected out of all neighbors in 6-digit postcode areas, without limiting the set of peers to similar individuals.[16] The corresponding RD estimate, which is also shown in Column (6) of Table 2, is small and insignificant. This confirms the main advantage of our IV strategy, which provides greater statistical power in detecting average peer effects in large networks. Second, Appendix Figure A4 reports a broader range of RD estimates using varying sets of characteristics in the peer assignment procedure. As in our sibling analysis, point estimates tend to increase in the number of characteristics used to match individuals with peers. Interestingly, adding gender and earnings does not seem to matter much once sector of work and migration background are included. We also note that spillover effects become more precisely estimated when we increase the number of matching characteristics. This occurs because in large groups such as neighborhoods, the number of unique pairs increases when using more variables to match individuals with peers, up to a point.

4.4 Retirement Spillovers in Workplaces

Several studies find that coworkers play an important role in shaping individual labor supply and career choices (e.g. Falk and Ichino, 2006; Mas and Moretti, 2009; Cornelissen et al., 2017). When it comes to retirement behavior, one might thus expect coworkers to be one of the most influential peer groups. Similar to neighborhoods, coworker networks tend to be large in our data. The median individual has 95 potential workplace peers. Therefore, we again implement both IV and RD strategies in order to address this challenge in two complementary ways.

Average Workplace Spillovers. We begin with the IV estimation leveraging the full variation in the reform exposure of workplace peers. Results are shown in Table 4. In Column (1), we find

[16] Note that while our main neighborhood definition is based on 5-digit postcodes, we use more fine-grained 6-digit postcodes for this specification in order to increase the likelihood of finding a unique neighbor without relying on additional characteristics.

a highly significant first-stage coefficient, implying that coworkers' leave-out average retirement age increases by 0.21 years per standard deviation of their reform exposure. Again, the first-stage F-statistic is very large. Column (2) shows a highly significant reduced-form spillover effect of 0.02 years. This corresponds to a small but precisely estimated scaled spillover effect of 0.10 years (1.3 months) per year of coworkers' later retirement. In Column (7), we run a placebo specification around an artificial reform cutoff in January 1948. The placebo test yields a small and insignificant estimate, indicating no correlation between individual retirement behavior and coworkers' local age structure, as required by our identification assumptions.

Influence of Specific Coworkers. As the second empirical strategy, we implement the RD design where we use the main socioeconomic characteristics observed in our administrative data in order to match individuals with their most similar workplace peer. We use the same set of variables as in the neighborhood analysis, namely gender, earnings, and an indicator for being foreign-born, plus city of residence, all measured at age 57. As before, we construct similar-coworker samples focusing on unique matches between the individual and a peer who share the maximum number of characteristics.[17] Panel (b) of Figure 4 shows our main RD results graphically. Among the most similar coworkers, we find a highly significant spillover effect of 0.11 years. Column (8) of Table 2 presents more details of this RD specification. A large first-stage effect of the 2006 reform on peers' retirement behavior occurs in the similar-coworker sample. When we scale the reduced-form estimate by the first stage, we find a spillover effect on individual retirement ages of 0.29 years (3.5 months) for each year of the coworker's later retirement. Thus, spillover effects between the most similar individuals are about three times larger than those between the average coworkers.

Spillovers by Firm Size. Coworker networks vary substantially in the number of peers, reflecting the large heterogeneity in firm size in our data. Panel (d) of Figure 4 investigates heterogeneity in spillovers by firm size by including interaction terms of coworkers' reform exposure with firm-size quintiles in our IV estimation. Detailed regression results are additionally shown in Table A1. We find that the largest retirement spillovers occur in small firms. In the bottom firm-size quintile, the scaled spillover effect is 0.21 years (2.5 months). In quintiles 2 to 4, scaled spillover effects are positive and significant at up to 0.14 years (1.7 months), while in the top quintile we detect no significant spillovers. These results are consistent with social interactions underlying the spillovers we document, as such interactions are likely more frequent in small firms (see Section 5 for a detailed discussion of these mechanisms).

[17] One issue faced by the workplace analysis is that the similar-coworker samples remain substantially smaller than the similar-neighbor samples, even when using the full set of characteristics in the matching procedure. This occurs mainly due to the skewness of the firm size distribution: many workers are employed in large firms where it is hard to identify unique peers. For this reason, we limit the sample to a slightly narrower set of birth cohorts around the reform cutoff, 1947 to 1952. This improves the chance of finding unique peers, increasing the number of observations available for the RD analysis.

Additional RD Results. Finally, our appendix contains two more RD results. First, Figure A3 shows an RD specification that uses only firm affiliation to select peers. This variant of the RD design effectively restricts the analysis to individuals who have a single coworker among the cohorts we study, which tends to occur in small firms. The corresponding point estimate of 0.14 years is large, which is consistent with our prior finding that spillovers are sizable especially in small firms. However, the estimate is somewhat imprecise and only marginally significant at the 10% level, again indicating that the IV design is better suited to detect broader coworker spillovers. Second, Appendix Figure A4 shows an extended set of RD results, varying the characteristics we use to assign peers. We find large and positive point estimates among all subsamples of coworkers sharing key socioeconomic characteristics. Similar to our results for neighbors, spillovers become more precisely estimated when more characteristics are used, as sample sizes increase.

4.5 Regression Discontinuity Design: Robustness and Specification Checks

We present a wide range of robustness and specification checks for all our main RD results in the Appendix. First, Appendix Figure A5 plots the distribution of individuals' own birth dates in the full sample, that is, the running variable of our first-stage estimation from Figure 1. Appendix Figure A6 similarly displays peers' birth date distributions, corresponding to the running variables in each of the samples we use to estimate spillover effects among spouses, siblings, neighbors, and coworkers in Figures 2, 3, and 4. We find no evidence of discontinuous patterns around the January 1950 cutoff in any of the estimation samples, supporting the corresponding RD identification assumption.

Second, Appendix Table A2 shows balancing tests using a set of pre-determined covariates, including gender, marital status, an indicator for being Dutch-born, and earnings, all measured at age 57. In addition, we include measures of network structure, in particular the number of siblings, coworkers, and neighbors, as outcomes. Both in the full sample and in the group-specific subsamples, we find that these characteristics do not exhibit any systematic discontinuities at the reform cutoff. In the few cases where the estimates are statistically significant, they remain very small in magnitude. These results further support the no-manipulation assumption underlying our RD design.

Third, a key assumption required for estimating spillover effects is the absence of sharp sorting by birth dates. As discussed in Section 3.1, such sorting would pose a problem for our identification strategy by confounding individuals' own reform exposure with that of their peers. We can test this assumption by plotting an indicator for own birth date being above the January 1950 cutoff by the respective peer's birth date. Appendix Figure A7 presents corresponding results for couples, siblings, neighbors, and coworkers. Across all estimation samples, the figure shows that individuals' own reform exposure evolves smoothly through the peer's birth date cutoff in January 1950. Even among couples, where one may perhaps expect age correlations to be of particular concern, sorting by birth date is not sharp enough to pose a problem for the RD design. A corresponding RD balancing test on own reform exposure is included in Appendix Table A2. Similarly, the table

shows that the Normal Retirement Age faced by individuals themselves exhibits no systematic discontinuities around the peer's cutoff across our RD samples.

Fourth, we examine the sensitivity of our RD results to key specification choices. Appendix Figure A8 displays both first-stage and spillover effects over a wide range of estimation bandwidths between 0.5 and 3 years, plotted in monthly increments. All our main RD estimates remain stable across bandwidths, except some spillover effects that become statistically insignificant when very narrow bandwidths are used. In addition, Appendix Table A3 reports more detailed results under alternative specifications, including a smaller (1-year) or a larger (3-year) bandwidth, a quadratic instead of a linear functional form, and a triangular instead of uniform kernel. Again, results remain very similar to our baseline estimation, corroborating the robustness of the RD design.

Finally, in order to further validate our empirical strategy, Appendix Figure A9 presents placebo RD results around an artificial cohort cutoff in January 1952, at which no policy change occurred. All other specification choices are identical to the main estimation. Across placebo specifications corresponding to all our main results, we find small and statistically insignificant placebo estimates. These placebo tests suggest that there are no other sources of correlation between individual retirement behavior and peers' birth dates that would bias our main results.

4.6 Taking Stock: Retirement Spillovers across the Social Network

Before we turn to mechanisms and policy implications, it is useful to take stock of our main empirical results so far. Our comprehensive analysis allows us to summarize and compare estimated spillover effects across multiple dimensions of the social network. Figure 5 shows our preferred set of estimates, combining RD estimates for spouses, siblings and similar neighbors and coworkers with IV estimates of the average neighbor and coworker effects. All spillover effects shown in the figure are scaled relative to the first stage in the respective sample, such that they are comparable in magnitude.

Our first key finding illustrated by Figure 5 is that many dimensions of the social network matter for individuals' retirement choices. We find that statistically and economically significant retirement spillovers arise in all social groups we study. Prior literature has documented these spillovers in couples, but has not systematically quantified spillovers between non-spousal peers, which our analysis reveals to be substantial.

Furthermore, the figure allows us to gauge the relative importance of different social relations for retirement decisions. Our comprehensive analysis is uniquely placed to enable such a comparison, since we examine family, neighborhood and workplace ties within a common setting. Spouses provide a natural benchmark: consistent with prior literature, spillovers in couples are generally large, and for women in particular the spouse is the most important social influence. As one may expect, the average non-spousal peer matters considerably less, exerting at most one-fifth of the average spousal effect. If individuals have very similar non-spousal peers, however, these can induce strong spillovers. Both for neighbors and coworkers, our estimates imply that the impact of similar peers is about three times larger than that of the average peer in these groups, reaching up

to 60% of the average spousal effect. Our point estimate for similar siblings likewise indicates a large spillover, though it is less precisely estimated given the smaller size of our sibling sample. As discussed previously, this similarity gradient is consistent with homophily in social interactions.

While prior studies offer little quantitative guidance on what magnitude of retirement spillovers one should expect between non-spousal peers, it is instructive to compare our findings to surveys of individuals' stated retirement preferences. In such surveys, individuals commonly report that their social network matters for their retirement plans. For instance, Vermeer et al. (2019) ask Dutch workers with whom they discuss retirement and whose choices they take into account for their own retirement decision. Respondents report interacting most with their spouse, followed by other family members, coworkers, and neighbors. This survey evidence corroborates our findings of which social relations matter for retirement behavior, with a ranking that is remarkably consistent with our main spillover estimates from Figure 5.

5 Mechanisms

Our main analysis uncovers significant retirement spillovers within families, neighborhoods, and workplaces. As we further discuss in Section 6, these estimates of total spillover effects across different peer groups are relevant for policy, as they govern the overall behavioral impacts and fiscal externalities of pension reforms. Nonetheless, it is informative to examine what mechanisms can explain the retirement spillovers we document. In this section, we investigate the role of three key potential mechanisms considered in the literature.

5.1 Information Spillovers

A first potential mechanism could be information spillovers. In our setting, changes in peers' retirement behavior are induced by the 2006 pension reform. Thus, spillover effects on other individuals could be driven by peers passing on information about the post-reform retirement rules, or individuals may learn about optimal retirement behavior under the new legislation from observing the choices of peers exposed to the reform. Crucially, this type of information should only be valuable for individuals who are subject to the post-reform rules, i.e. individuals who are themselves born in January 1950 or later. Hence, in order to test this information mechanism, we investigate to what extent spillovers are concentrated among relatively young individuals who are themselves exposed to the reform.

For spouses and siblings, we estimate RD specifications like our main ones, splitting the sample into individuals born before vs. after January 1950 (see Appendix Figure A10). For neighbors and coworkers, we perform an analogous exercise within our IV framework, including an interaction term between the fraction of peers exposed to the reform and an indicator for the individual being born above the January 1950 cutoff. The latter results are shown in Column (3) of Tables 3 and 4, respectively. We find that retirement spillovers in couples are larger among women who are themselves exposed to the reform. However, the opposite pattern emerges for siblings, neighbors,

and coworkers. In those groups, spillover effects of a peer's reform exposure are larger among individuals born before January 1950 – who are not subject to the post-reform rules – than among younger individuals. The difference is particularly substantial among coworkers, where the scaled spillover effect is 0.25 years (3.0 months) for non-exposed individuals, while the point estimate is insignificant for exposed individuals.

We conclude from this evidence that an information mechanism is unlikely to explain our results. Among siblings, neighbors and coworkers, spillover effects are not driven by individuals for whom information about the post-reform retirement rules would be relevant. If anything, we find larger spillovers among older individuals who face a different set of rules. As we discuss below, this is consistent with the propagation of social norms about retirement rather than information. The only exception is given by spouses, where spillovers are strong among younger women. Yet it is not clear whether this specific pattern can be interpreted as an effect of reform exposure per se; couples' relative age could be correlated with other factors such as gender attitudes that shape how strongly women respond to their husbands' retirement choices.

5.2 Leisure Complementarity

A second potential mechanism behind retirement spillovers could be leisure complementarity. That is, individuals may place higher value on leisure when it is spent in the company of others. If an important peer retires later, the individual may thus decide to postpone retirement, as the value of leisure is lower in the meantime. There is evidence in favor of leisure complementarity in particular within couples (e.g. Georges-Kot et al., 2024). In the context of retirement, the empirical signature of leisure complementarity typically examined in the literature is individuals retiring at the same time (Garcia-Miralles and Leganza, 2024).

Hence, we investigate to what extent joint retirements of individuals with their spouse or with other peers can explain our main results. We run RD specifications analogous to Figures 2, 3, and 4, splitting each sample into pairs retiring jointly vs. pairs retiring separately. To generously capture potential leisure complementarity, we define joint retirements as cases where the individual retires within +/- 6 months of the peer's retirement date; separate retirements are those occurring outside this window. Figure 6 summarizes the results in a coefficient plot, and the full set of corresponding RD plots are shown in Appendix Figure A11. Indeed, we find that spillover effects are generally larger among pairs retiring jointly than among those retiring separately. This pattern holds across all peer groups except siblings, where the estimate for joint retirements is noisy due to the small sample size. Importantly, however, spillovers remain sizable and highly significant among pairs who retire separately across all groups.

Furthermore, it is important to note that the fraction of joint retirements is limited across all types of peers. Even in couples, where prior literature suggests an important role for joint retirement, only about 15% of pairs retire within +/-6 months of each other in our data. Among the most similar siblings, neighbors and coworkers, the corresponding shares are 7%, 7%, and 12%, respectively. Overall, these results suggest that leisure complementarity can generate strong re-

tirement spillovers for a subset of individuals. Joint retirements occur most frequently in couples, and to a lesser extent among similar coworkers and neighbors. However, given that only a small fraction of pairs retire jointly, and that spillovers also arise among individuals whose retirement dates are far apart, this mechanism is unlikely to account for the majority of spillover effects.

5.3 Social Norms

A third potential mechanism behind our results could be that social norms about retirement are transmitted via peer networks. A growing body of literature documents retirement patterns that are hard to reconcile with standard models, but are consistent with norm-driven behavior. In particular, a well-documented empirical fact is that many workers retire exactly at the Normal Retirement Age (NRA), as they perceive this age as a reference point – a "normal" time to retire (Behaghel and Blau, 2012; Gruber et al., 2022; Lalive et al., 2023; Seibold, 2021, 2024; Reck and Seibold, 2026). Similar to other countries, this behavior is prevalent in the Netherlands, and prior work points to social norms as a mechanism underlying it (van Erp et al., 2014; Vermeer et al., 2019; Rabaté et al., 2024).[18] In the same vein, we observe a strong concentration of retirements precisely at the NRA in Panel (b) of Figure 1, even though there are no financial retirement incentives linked to this age. Social interactions may be an important factor in determining whether workers adopt such notions of normal behavior.

The 2006 reform provides a unique opportunity to test how social norms about retirement are propagated. As we pointed out in Section 4.1, one side effect of the reform is a substantial increase in the retirement spike at the NRA, as exposed individuals retire later. Panel (a) of Figure 7 more formally quantifies the impact of the reform on the probability of bunching at the NRA in our full sample. We estimate a first-stage RD specification akin to equation (1), replacing the individual's retirement age by their probability to retire at the NRA (defined as retirements within +/- 1 month of the NRA) as the outcome. Among workers born just below the January 1950 cutoff, the baseline fraction retiring at the NRA is around 10%. The RD estimate implies an increase in the probability of bunching at the NRA of 5.6 percentage points, corresponding to an increase of more than 50% in the prevalence of this behavior.

We then exploit this strong first stage to study whether individuals' propensity to retire at the NRA is influenced by their peers' behavior. Panel (b) of Figure 7 summarizes estimated spillover effects, scaled relative to the corresponding first stage. For spouses and siblings, we estimate a variant of the RD design from equation (2) using the probability of bunching at the NRA as the outcome. We find significant spillovers in couples: women become 1.9 percentage points more likely to retire at the NRA when their husband is exposed to the reform. This effect is large, corresponding to 32% of the first-stage increase in the probability of bunching at the NRA. Among similar

[18] Besides norms, another potential reason for retiring at the NRA in the Dutch setting could be automatic contract termination (Rabaté et al., 2024). While such a firm-side mechanism could account for some of the observed bunching at the NRA, it is unlikely to explain the spillovers we measure in this section for two reasons. First, firms' scope to terminate workers at the NRA is unaffected by the 2006 reform. Second, we find spillovers in bunching at the NRA not only among coworkers, but also for family members and neighbors, for whom firm-side incentives should play no role.

siblings, we find no significant spillover effect in terms of retiring at the NRA. This specification is estimated on a considerably smaller sample, however, and the confidence interval includes effect sizes we estimate for other non-spousal peers below.

For neighbors, we turn to our IV framework. We estimate equations (4) and (5), replacing the individual's retirement age by the individual's probability to retire at the NRA, and peers' leave-out average retirement age by the leave-out share of peers retiring at the NRA. Column (4) of Table 3 shows a highly significant first-stage estimate, implying that the fraction of neighbors retiring at the NRA increases by 1.1 percentage points per standard deviation of neighbors' reform exposure. Our main spillover effect estimate in Column (5) is positive and significant, and corresponds to a scaled spillover effect of around 13% of the first stage. This estimate is also displayed in Panel (b) of Figure 7 for comparison to the other peer groups. Reassuringly, in Column (8), a placebo specification around an artificial reform cutoff yields no significant correlation between the probability of retiring at the NRA and neighbors' cohort composition.

We proceed analogously for coworkers. In Column (4) of Table 4, we find a strong first-stage effect of workplace peers' reform exposure on the leave-out fraction bunching at the NRA. Column (5) shows a small but significant spillover effect on the individual's likelihood of retiring at the NRA. The scaled spillover effect amounts to 10% of the first stage. A placebo specification again yields no significant effect, supporting our IV identification assumptions.

These findings suggest that part of the retirement spillovers documented by our main analysis reflect norm-related behavioral patterns propagating through social networks. One additional piece of evidence further supports this interpretation. Column (6) of Tables 3 and 4 shows that both among neighbors and coworkers, spillovers in bunching at the NRA are largely concentrated among individuals who are not themselves exposed to the reform. In other words, the reaction to peers retiring at the NRA comes from individuals who can retire under the pre-reform rules and therefore face no incentive to shift their own retirement towards the NRA. This mirrors the general pattern from Section 5.1 where spillovers generally run from exposed to non-exposed individuals.

5.4 Other Mechanisms

Financial Incentives in Couples. A specific channel that could matter for our analysis of couples is financial linkages between spouses. As the 2006 reform lowers exposed individuals' pension wealth, their spouse may increase labor supply due to a wealth effect on the household budget constraint. If this mechanism were at play, we would expect spousal responses to increase in the size of the pension wealth shock. A key source of variation in the size of the financial shock is whether the household's primary or secondary earner is exposed to the reform. We investigate this in Appendix Figure A2, where we analyze spillovers by relative earnings. We find no significant differences in spillovers by relative earnings: women respond to their husband's reform exposure in a similar way across primary- and secondary-earner husbands, and men do not respond in either case. This suggests that the shared household budget constraint is not the main driver of our results.

A related point concerns the partner allowance, which exists during part of our analysis period and may create financial incentives for younger spouses to retire as soon as the older spouse reaches the NRA (see Section 2.2). This policy is unlikely to explain our results for couples for a few reasons. First, it only affects couples in which the older partner retires exactly at the NRA, which, despite bunching at the NRA, is a limited share of our sample. Second, the financial incentive applies to younger, secondary-earner spouses regardless of gender, but we find spillovers concentrated among women regardless of these characteristics. Third, if the partner allowance was responsible for joint retirement, the probability of this behavior should drop after its removal, which affects couples whose older spouse is born in 1950 (Nagore Garcia and van Soest, 2022). Empirically, however, the probability of joint retirement remains remarkably stable throughout our RD estimation bandwidth, with no evidence of any discontinuous change (see Appendix Figure A13).

Firm Responses. For our analysis of coworkers, firm responses could play a role. While labor demand can be an important determinant of retirement behavior, two pieces of evidence suggest that firm responses are unlikely to account for the spillovers we observe between coworkers. First, if old workers are substitutes from firms' point of view, as much of the literature finds (e.g. Bianchi et al., 2023; Jäger et al., 2024), one would expect negative employment spillovers between them, i.e. a coworker's later retirement should reduce demand for other old workers. However, we consistently find positive retirement spillovers between coworkers, and these spillovers become larger for more similar coworkers who, if anything, should be more substitutable for firms.

Second, firms with more workers exposed to the 2006 reform may change their internal retirement policies for other old workers, for instance because they apply uniform human-resource policies. Heterogeneity in spillovers by firm size is informative about this mechanism, as larger firms are more likely to have structured, formalized people-management practices (Bloom and Van Reenen, 2010). In Figure 4, we find that retirement spillovers are larger in small firms, however. As discussed previously, this speaks in favor of social interactions in the workplace, rather than firms' retirement policies, driving our results.

6 Policy Implications

The key policy implication of our findings is that social spillovers exacerbate the overall impact of pension reforms on retirement behavior. Building on our comprehensive analysis across multiple peer groups, we can aggregate these effects and quantify the *social multiplier* of pension reform. This calculation yields three key insights that extend beyond the analysis of specific peer groups. First, it allows us to gauge the overall quantitative importance of social interactions in retirement behavior, which is crucial for policymakers interested in the impact of pension reforms on a population. Second, as the literature on optimal social insurance points out, the total behavioral response to pension reforms is a key input determining their fiscal and welfare effects

(Kolsrud et al., 2024). Our multiplier calculation provides a measure of how social spillovers modify this total behavioral response.[19] Third and related, the social multiplier is informative about the degree to which standard individual-level studies that do not account for social spillovers may underestimate the total effect of pension reforms.

Underlying our notion of a social multiplier is a thought experiment analogous to the 2006 pension reform. Suppose a reform targets some individuals in a population and, through its direct effect on the behavior of these exposed individuals, leads to some change in the average retirement age. We then define the social multiplier as the factor by which social spillovers amplify the impact of the reform on the average retirement age in this population, relative to its direct effect (cf. Glaeser et al., 2003).

We calculate the social multiplier M as:

$$M = 1 + \sum_k \overline{N}_{-i,k} \sum_{\theta \in \Theta_k} p_\theta \mu_\theta \tag{6}$$

where k indexes peer groups (spouses, siblings, neighbors, and coworkers) and θ indexes types of peers within a group, for example male vs. female spouses or siblings with varying degrees of similarity. $\overline{N}_{-i,k}$ is the average number of peers within group k, Θ_k is the set of types in k, and p_θ is the fraction of type θ in the group. μ_θ denotes the estimated spillover effect for peers of type θ, scaled by the direct effect of the reform. For spillover effects estimated through our RD framework, μ_θ corresponds to the coefficient β_1 from equation (2) divided by the first-stage coefficient α_1. In the IV framework, μ_θ corresponds to δ_1 from equation (5) divided by the first-stage effect ϕ_1.

Our preferred specification operationalizes equation (6) as follows. For couples, we combine the RD estimates from Figure 2 with the fraction of individuals with a working spouse and the gender composition of our sample. For siblings, we use the set of estimates from Panel (c) of Figure 3 together with the average number of siblings and the relative sizes of the subsamples with different degrees of similarity.[20] For neighbors and coworkers, we directly use our IV estimates from Tables 3 and 4, as these correspond to the average impact of neighborhood or workplace peers.

It is useful to clarify a few assumptions and choices on which this implementation rests. In our baseline specification, we make these choices conservatively, staying in line as closely as possible with our empirical analysis and minimizing extrapolation of our estimates to other peers. First, we restrict attention to peers among the cohorts included in our estimation samples, namely those born between 1946 and 1954. This is conservative, as it disregards any spillovers between the same types of peers outside this window. Second, when we use subgroup estimates, we take weights $\overline{N}_{-i,k}$ and p_θ directly from our estimation samples, so that spillovers are applied only to peers for whom we estimate them, with no extrapolation to different types of peers. Third,

[19] Kroft (2008) formalizes the argument that a social multiplier alters optimal social insurance provision in the context of unemployment benefits.

[20] In order to increase precision in the sibling component of the multiplier, our main specification focuses on those subsamples where we find significant spillover effects.

when we use IV estimates, we incorporate them based on the same linear-in-means specification as in the empirical analysis, assuming that the influence of a given peer decreases with group size. Fourth, we carefully check for network overlap and ensure our multiplier is not inflated by double-counting of peers.[21] Finally, while we do not explicitly model potential interactions of spillovers across peer groups, we note that these are accounted for by summing over estimated average effects.[22] We further discuss robustness to varying some of these choices below.

Main Results. Figure 8 illustrates the results of the multiplier calculation for our main outcome, the average retirement age. More detailed results are shown in Table 5. We find a social multiplier of 1.42, with a standard error of 0.06. Thus, for a pension reform inducing a one-year increase in the average retirement age through its direct effect, the overall impact on the average retirement age is 1.42 years. In other words, social spillovers strongly amplify the behavioral impact of the reform. Out of the total spillover effect of 42%, spouses account for 19 percentage points and siblings account for 7 percentage points. Neighbors and coworkers contribute 6 and 10 percentage points, respectively. While spouses are the largest individual contributors, non-spousal peers thus account for more than half (23 percentage points) of the total multiplier, underscoring the importance of incorporating them into the analysis of pension reforms.

Table 5 also reports the social multiplier on two additional outcomes, total earnings in the population and the fiscal impact of the pension reform. To compute these measures, we use our estimated spillover effects together with information on monthly earnings in each subsample, and then employ a tax simulator to quantify government revenue associated with delayed retirement.[23] The resulting social multiplier is 1.35 for total earnings and 1.34 for the fiscal impact. Hence, social spillovers substantially exacerbate fiscal effects of the reform, which are often a central policy objective. The slightly smaller magnitude relative to the multiplier for the average retirement age reflects the gender asymmetry we document among couples. Because spillovers are concentrated among women and these have lower average earnings, their later retirement entails a smaller impact on total earnings, and in turn on fiscal revenues. In contrast, the group-specific multipliers for siblings, neighbors and coworkers are very similar across the different outcomes.

[21] We do so in three ways. First, subgroups within type k are always defined in a mutually exclusive way. Second, we exclude spouses from all other peer samples, such that spouses are not also counted as neighbors or coworkers. Third, we empirically check for network overlap among non-spousal peers and find that sibling, neighbor and coworker networks are virtually fully distinct. In our data, only 0.24% of siblings are also neighbors and 0.54% are coworkers; 0.13% of neighbors are also siblings and 0.75% are coworkers; 0.39% of coworkers are also siblings and 0.59% are neighbors.

[22] For example, if siblings exert larger spillovers for individuals who do not have a spouse, this is captured through the estimated average sibling effect, which is a weighted average across married and unmarried individuals.

[23] In our setting, fiscal externalities from delayed retirement occur due to additional income tax receipts and social insurance contributions. We simulate government revenue from these sources arising both from the direct effect and from spillover effects in each subsample. There are no fiscal effects from delayed pension benefit receipt because second-pillar pensions feature actuarially fair benefit adjustment after the 2006 reform, and thus later retirement does not affect their present discounted value. Moreover, the reform does not change the timing of first-pillar pension claims.

Robustness. As discussed above, calculating the social multiplier requires some choices about what estimates to incorporate and how to define the relevant set of peers whose behavior is modified by spillovers. To examine the sensitivity of our results to these choices, Appendix Table A4 reports results under a range of alternative specifications. Using only average spillover effects rather than subgroup estimates yields a slightly larger multiplier on the average retirement age of 1.45. While our preferred specification combines RD estimates for spouses and siblings with IV estimates for neighbors and coworkers, relying exclusively on RD estimates for all peer groups produces a slightly smaller multiplier of 1.35. We also consider specifications that relax the assumption that spillovers only occur within our estimation sample. Expanding the birth cohort window to 1942 to 1954 (the set of cohorts for whom we observe the required range of retirement ages) increases the multiplier to 1.49, and additionally relaxing the work restriction on peers raises the multiplier to 1.62. Overall, these results indicate that the social multiplier remains quite similar or becomes even larger under alternative specification choices.

Interpretation. In general, our social multiplier calculation is of course predicated on the characteristics of our sample and the type of variation arising from the Dutch 2006 reform. For instance, female labor force participation and gender norms may change across cohorts, which may modify the couples component of the multiplier. Within our setting, our multiplier calculation likely provides a lower bound on how social interactions affect overall retirement responses to pension reforms. First, although our analysis covers the types of peers considered in much of the literature on peer effects, there could in principle be further unobserved peer groups causing retirement spillovers, such as friends without any formal ties to the individual. As long as average spillover effects are positive within additional groups, which seems plausible given that we find positive effects across all types of peers we observe, these would add to the social multiplier of a pension reform. Second, we only focus on "first-round" spillover effects and do not take into account potential cascading dynamics. If individuals subject to peer effects subsequently exert retirement spillovers on others, this would again exacerbate the multiplier (see e.g. Dahl et al., 2014b).

7 Conclusion

In this paper, we provide novel empirical evidence on the role of social forces for retirement behavior. We find large retirement spillovers in couples, which mainly occur due to women reacting to their husband's choices. While the influence of the average sibling, neighbor and coworker on individual retirement behavior is small, we estimate sizable spillover effects within all these groups when individuals are similar in terms of sociodemographic characteristics. These results are consistent with the notion of homophily in social networks, which posits that similar individuals are more likely to interact and influence each other's behavior. Furthermore, we provide evidence that retirement spillovers are driven both by leisure complementarities and by the transmission of social norms.

Our findings have important implications for pension policy. Since a change in an individual's retirement age influences others in their social network, there is a *social multiplier* effect that amplifies the overall impact of pension reforms on retirement behavior. Not taking into account this multiplier would lead researchers and policymakers to underestimate the total labor market and fiscal effects of reforms. Furthermore, our findings imply that social spillovers can significantly alter the welfare impact of policies. Fiscal externalities arising from behavioral responses are a key input into welfare evaluations of pension reforms, as they reflect the moral hazard cost of additional government spending on pensions (Kolsrud et al., 2024). For a given value of benefits to recipients, optimal pension system generosity may be lower when taking into account that social spillovers amplify fiscal externalities. Similarly, the marginal value of public funds (MVPF) of pension reforms depends on total behavioral responses (Hendren and Sprung-Keyser, 2020), and incorporating our social multiplier would lower the MVPF of extra spending on pension benefits.

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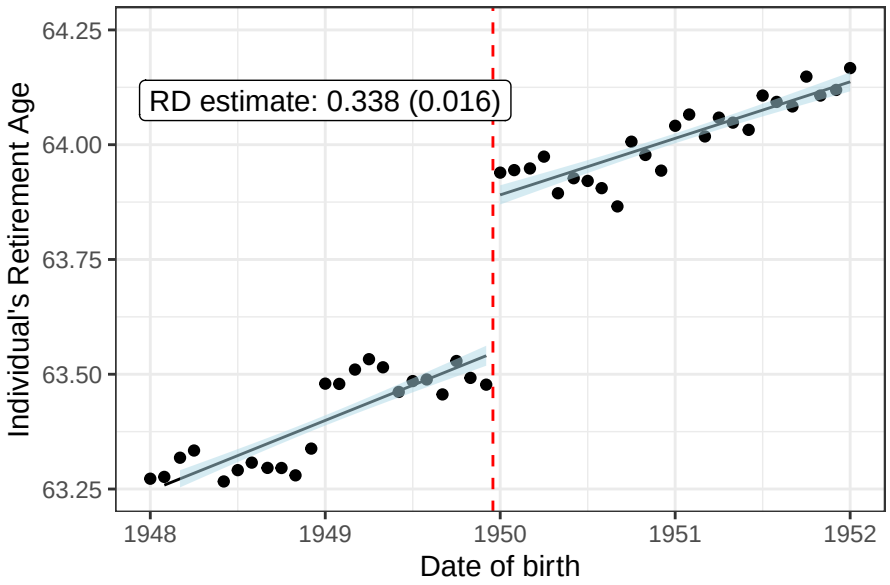
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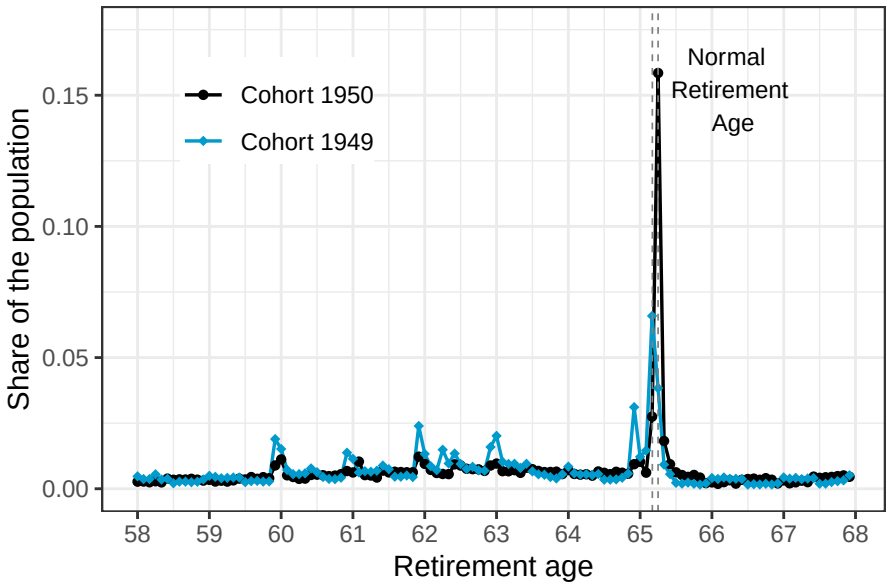
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FIGURE 1: DIRECT EFFECT OF THE 2006 REFORM

(a) Average Retirement Age



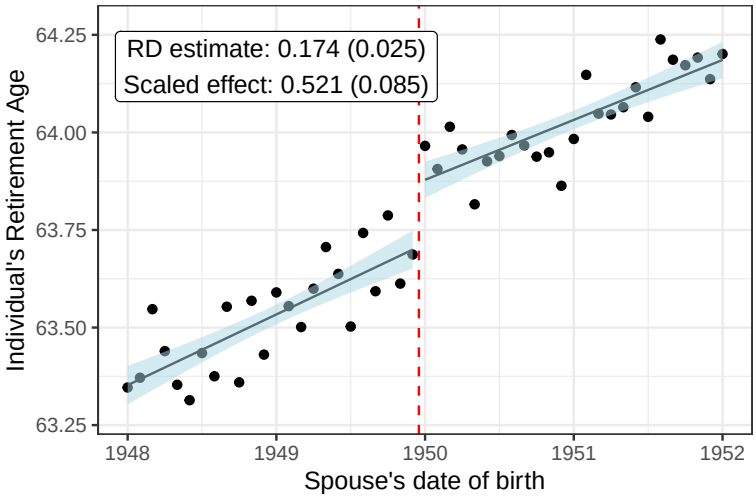
(b) Retirement Age Distribution



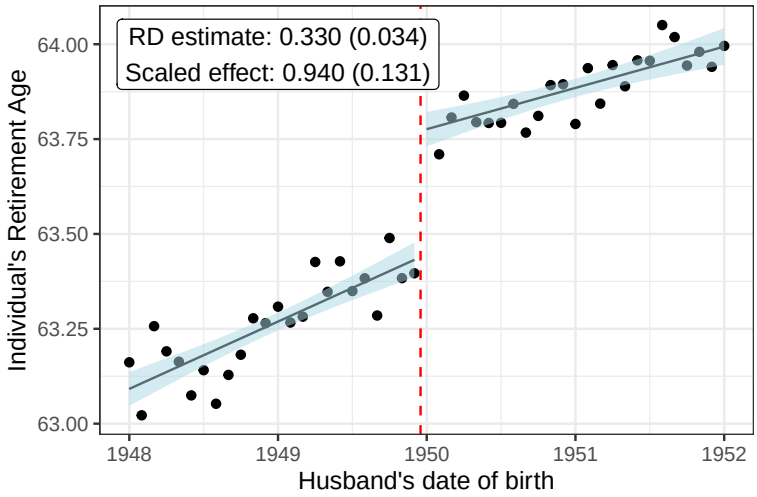
Notes: Panel (a) of the figure shows individuals' average retirement ages by their own birth date in monthly bins. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses. Panel (b) shows the distribution of retirement ages among individuals born in 1949, the last year below the reform cutoff, and among individuals born in 1950, the first year above the reform cutoff. The dashed vertical lines demarcate the Normal Retirement Age, which is 65 years and 2 months for most individuals born in 1949 and 65 years and 3 months for most individuals born in 1950.

FIGURE 2: SPILLOVERS IN COUPLES

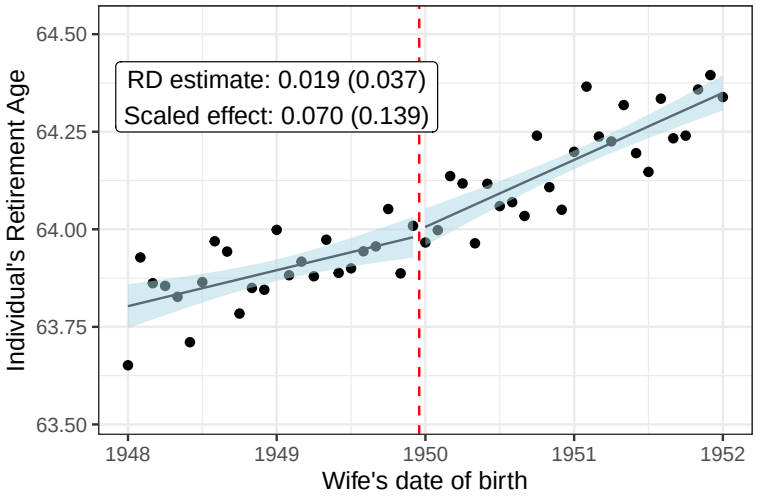
(a) All Spouses



(b) Women

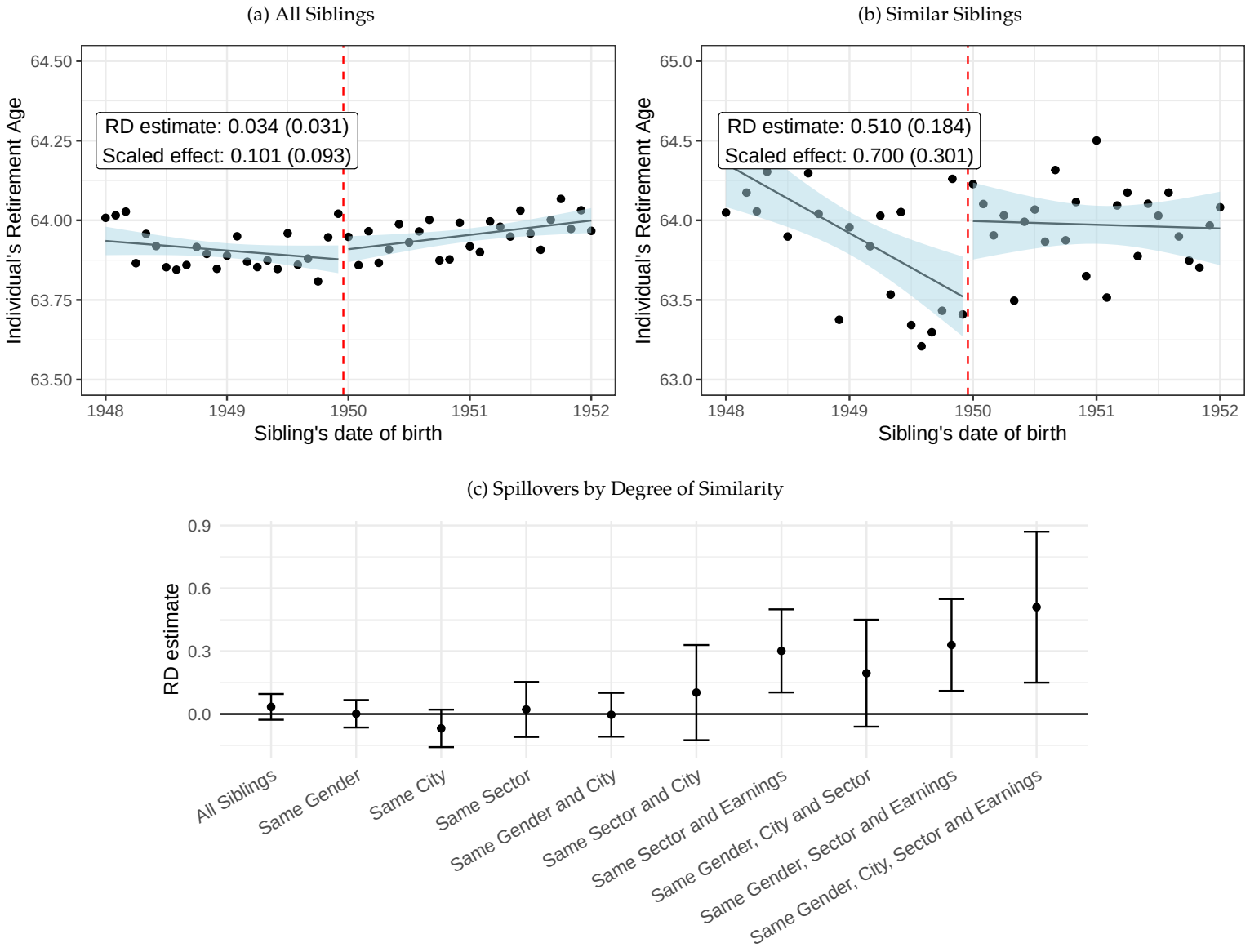


(c) Men



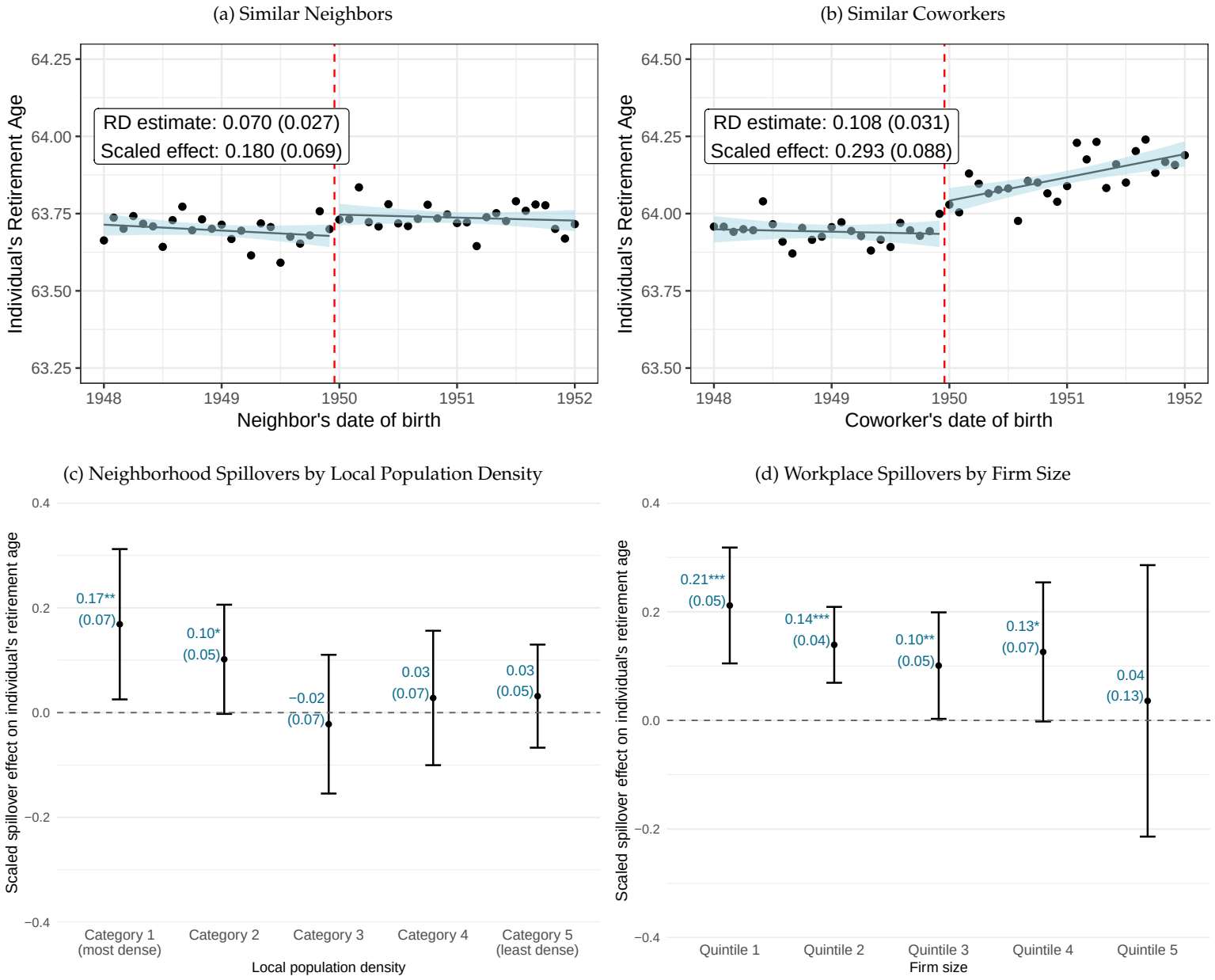
Notes: The figure shows individuals' average retirement ages by their spouse's birth date in monthly bins. Panel (a) pools all spouses, Panel (b) focuses on women and Panel (c) focuses on men. In each panel, the red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. In each panel, the text box displays the reduced-form RD estimate of the jump in retirement ages at the cutoff as well as the implied scaled spillover effect, with standard errors in parentheses.

FIGURE 3: SPILLOVERS BETWEEN SIBLINGS



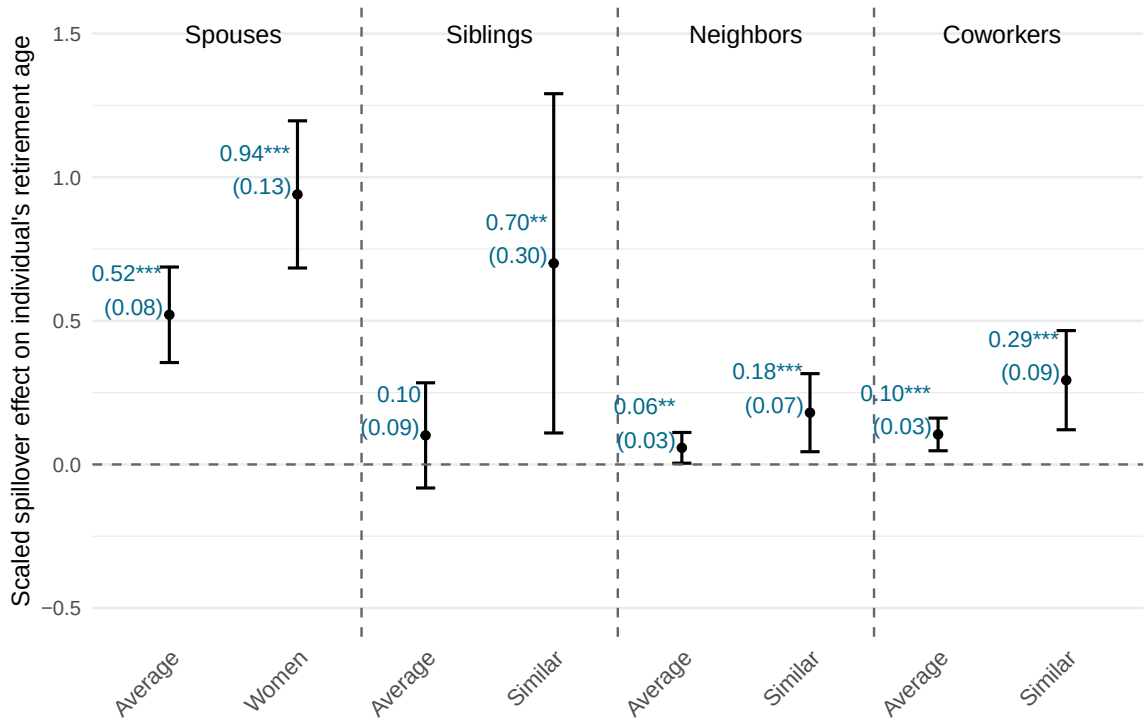
Notes: Panels (a) and (b) of the figure show individuals' average retirement ages by their sibling's birth date in monthly bins. Panel (a) includes all siblings and Panel (b) includes similar siblings who have the same gender, live in the same city, work in the same 2-digit sector, and are in the same earnings decile. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. In both panels, the text box displays the reduced-form RD estimate of the jump in retirement ages at the cutoff as well as the implied scaled spillover effect, with standard errors in parentheses. Panel (c) shows an extended set of RD estimates with 95% confidence intervals for subsamples of siblings indicated by the labels on the horizontal axis.

FIGURE 4: SPILLOVERS BETWEEN NEIGHBORS AND COWORKERS



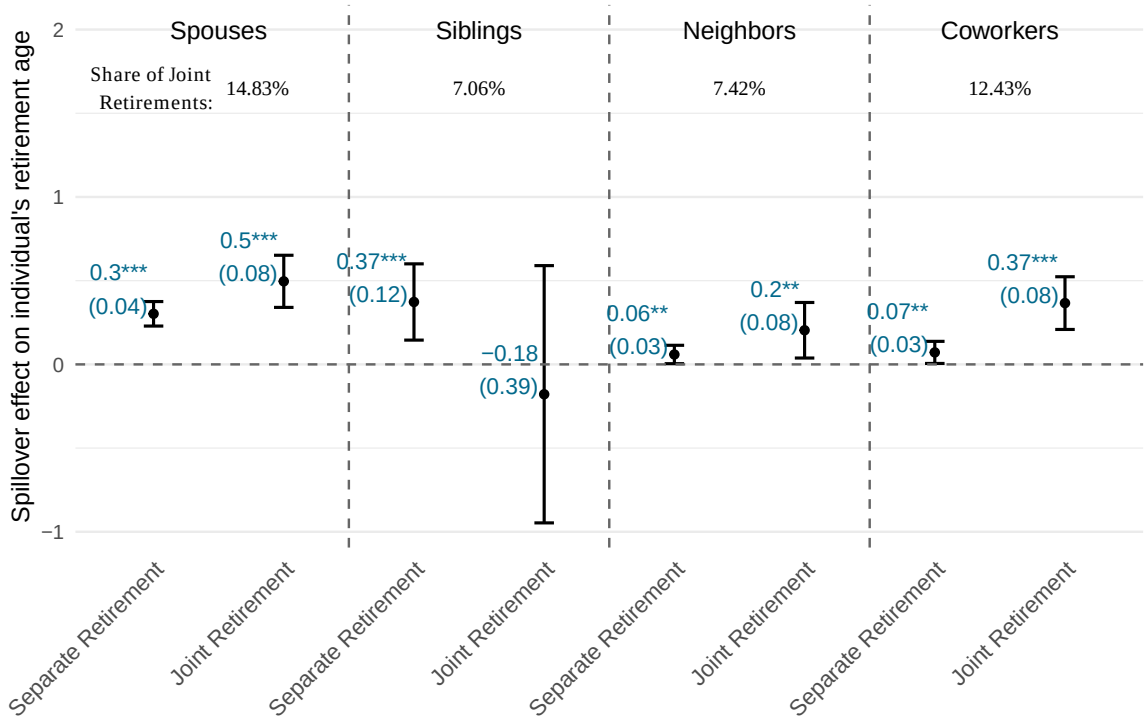
Notes: The figure shows retirement spillovers between neighbors and coworkers. Panels (a) and (b) display individuals' average retirement ages by their neighbor's or coworker's birth date in monthly bins. Panel (a) uses individuals' most similar neighbors based on 2-digit sector of work, an indicator for being foreign-born, gender, and earnings decile as peers. Panel (b) uses individuals' most similar coworkers based on an indicator for being foreign-born, gender, city of residence and earnings decile as peers. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. The text boxes display the reduced-form RD estimate of the jump in retirement ages at the cutoff as well as the implied scaled spillover effect, with standard errors in parentheses. An extended set of RD estimates for neighbors and coworkers are shown in Appendix Figure A4. Panels (c) and (d) show heterogeneous spillover effects of neighbors' and coworkers' reform exposure. Results are obtained via the instrumental variables approach, interacting peers' reform exposure with indicators for density categories (Panel c) and firm size quintiles (Panel d). In Panel (c), the following density categories are used: 1 is extremely urbanized (more than 2500 addresses per square kilometer), 2 is strongly urbanized (1501 to 2500 addresses), 3 is moderately urbanized (1001 to 1500 addresses), 4 is hardly urbanized (501 to 1000 addresses), 5 is not urbanized (500 or fewer addresses). In both panels, each point corresponds to an estimated scaled spillover effect with its 95% confidence interval. See Appendix Table A1 for detailed regression estimates.

FIGURE 5: COMPARING SPILLOVERS ACROSS THE SOCIAL NETWORK



Notes: The figure summarizes our main estimates of retirement spillovers across the social network. The effects for spouses, siblings, similar neighbors and similar coworkers are regression discontinuity estimates, see Table 2 for details. The average neighbor and coworker effects are instrumental variable estimates, see Tables 3 and 4 for details. Each point corresponds to an estimated scaled spillover effect with its 95% confidence interval.

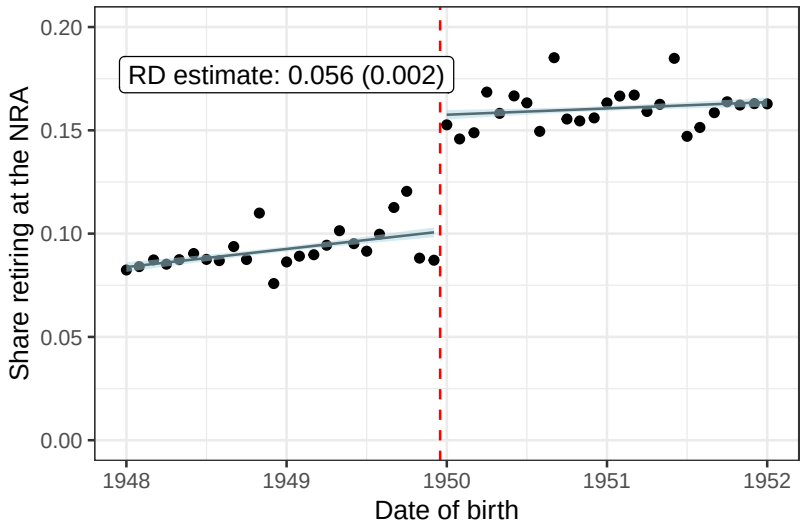
FIGURE 6: THE ROLE OF JOINT RETIREMENT



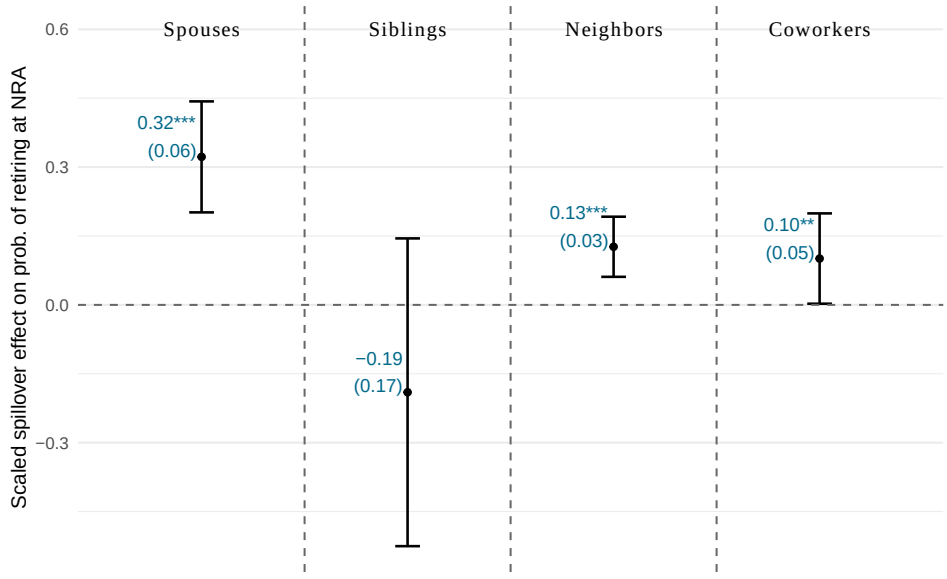
Notes: The figure shows retirement spillovers among pairs retiring jointly vs. retiring separately. The results for all peer groups are obtained via a regression discontinuity design, where the sample is split into individuals retiring within +/- 6 months of their peer's retirement date vs. individuals retiring outside of this window. See Appendix Figure A11 for full RD results. Each point corresponds to an estimated spillover effect with its 95% confidence interval.

FIGURE 7: PROPAGATION OF SOCIAL NORMS

(a) First Stage (Full Sample)

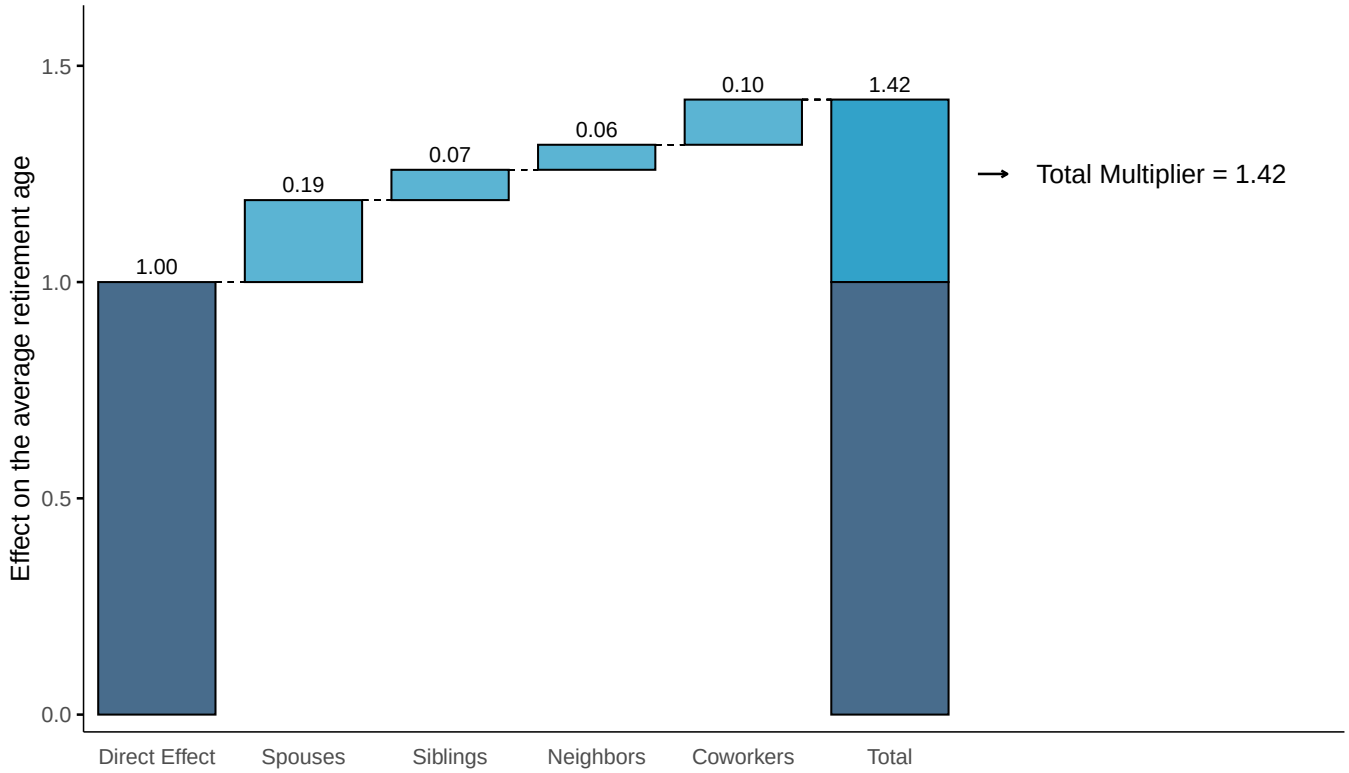


(b) Spillover Effects



Notes: Panel (a) of the figure shows individuals' probability of retiring exactly at the Normal Retirement Age by their own birth date in monthly bins. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An RD estimate of the jump in the share retiring at the NRA at the cutoff is included in the text box, with its standard error in parentheses. Panel (b) of the figure shows estimated spillover effects on the probability of retiring exactly at the NRA when a peer is exposed to the reform. For spouses and siblings, the results are obtained via a regression discontinuity design. See Appendix Figure A12 for full RD results. The results for neighbors and coworkers are obtained via the instrumental variable approach, using the probability of retiring at the NRA as the regression outcome. The IV results shown here are based on the estimates from Column (5) of Table 3 and Column (5) of Table 4. Each point corresponds to an estimated scaled spillover effect with its 95% confidence interval.

FIGURE 8: THE SOCIAL MULTIPLIER OF PENSION REFORM



Notes: The figure illustrates how we calculate the social multiplier of a pension reform. As the first bar on the left shows, we consider a pension reform with a direct effect on the average retirement age normalized to one. The next four bars depict group-specific multipliers for spouses, siblings, neighbors, and coworkers, quantifying each group's contribution to the total multiplier. The last bar on the right illustrates the total social multiplier aggregated across all peer groups.

TABLE 1: SUMMARY STATISTICS

	(1) Mean	(2) S.D.	(3) Median
Panel A: Full sample (N=1,367,064)			
Female	0.39	0.49	0.00
Married of civil partnership	0.76	0.43	1.00
Dutch-born	0.86	0.35	1.00
Annual labor income at 57	34915.04	42893.15	30416.00
Year of birth	1950.52	2.62	1950.58
Birth date $\geq$ January 1950	0.57	0.50	1.00
Retirement age	63.81	3.16	64.42
Retirement at the NRA	0.16	0.37	0.00
Retirement before the NRA	0.60	0.49	1.00
Panel B: Couples Sample (N=479,418)			
Number of spouses	1.00	0.00	1.00
Spouses' year of birth	1950.58	2.46	1950.58
Spouses' retirement age	63.89	3.06	64.33
Panel C: Siblings Sample (N=604,776)			
Number of siblings	1.69	0.95	1.00
Siblings' year of birth	1950.73	2.16	1950.75
Siblings' retirement age	63.97	2.70	64.25
Panel D: Neighbors Sample (N=1,326,820)			
Number of neighbors	61.62	32.77	57.00
Neighbors' year of birth	1950.59	0.57	1950.58
Neighbors' retirement age	63.86	0.62	63.85
Panel E: Coworkers Sample (N=928,842)			
Number of coworkers	539.23	1449.36	95.00
Coworkers' year of birth	1950.70	1.54	1950.81
Coworkers' retirement age	64.12	1.19	64.14

Notes: The table shows summary statistics of our data. Panel A shows individual characteristics in the full sample. Panels B to E describe the characteristics of peers in the subsamples we use to analyze the respective groups indicated in the panel titles. The number of observations (denoted by N) in each of the subsamples in Panels B to E corresponds to the number of individuals with at least one peer of the respective type that satisfies our sample inclusion criteria. "S.D." denotes standard deviation.

TABLE 2: REGRESSION DISCONTINUITY ESTIMATES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Couples		Siblings		Neighbors		Coworkers	
	All	Men	Women	All	Similar	All	Similar	All	Similar
Panel A: First Stage				*Outcome: Peer's Retirement Age*					
Peer's birth date above 1950 cutoff	0.333^{***}	0.270^{***}	0.351^{***}	0.338^{***}	0.728^{***}	0.315^{***}	0.391^{***}	0.340^{***}	0.369^{***}
	(0.025)	(0.036)	(0.033)	(0.030)	(0.171)	(0.058)	(0.025)	(0.071)	(0.031)
Panel B: Spillover Effect (Reduced Form)				*Outcome: Individual Retirement Age*					
Peer's birth date above 1950 cutoff	0.174^{***}	0.019	0.330^{***}	0.034	0.510^{***}	-0.071	0.070^{***}	0.136^{*}	0.108^{***}
	(0.025)	(0.037)	(0.034)	(0.031)	(0.184)	(0.061)	(0.027)	(0.072)	(0.031)
Panel C: Spillover Effect (Scaled by First Stage)				*Outcome: Individual Retirement Age*					
Peer's birth date above 1950 cutoff	0.521^{***}	0.070	0.940^{***}	0.101	0.700^{**}	-0.225	0.180^{***}	0.401^{*}	0.293^{***}
	(0.085)	(0.139)	(0.131)	(0.093)	(0.301)	(0.198)	(0.069)	(0.227)	(0.088)
Observations in bandwidth	232,244	108,448	123,810	160,889	4,104	44,565	212,323	24,316	112,307
F-statistic (first stage)	182.82	55.21	114.96	127.80	18.08	29.19	239.47	22.92	144.73

Notes: The table summarizes regression discontinuity estimates across all peer groups. Column titles indicate the respective type of peer. Panel A shows first-stage results for each sample. The coefficients shown are obtained from an RD design, estimating the impact of the respective peer's reform exposure on their own retirement age. Panel B displays spillover effects obtained from an RD design estimating the impact of the peer's reform exposure on the individual's retirement age. The estimates in Panel B correspond to the reduced-form spillover effects depicted in Figures 2 to 4. Panel C shows spillover effects scaled relative to the respective first-stage effect. Standard errors are in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

TABLE 3: INSTRUMENTAL VARIABLE ESTIMATES FOR NEIGHBORS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Main Results			Social Norms			Placebo Results	
	First Stage: Neighbors' retirement age	*Spillover:* Individual retirement age		*First Stage:* Fraction of neighbors retiring at NRA	*Spillover:* Probability to retire at NRA		*Spillover:* Individual retirement age	*Spillover:* Probability to retire at NRA
Neighbors' reform exposure (standardized)	0.1162^{***}	0.0067^{**}	0.0090^{*}	0.0108^{***}	0.0014^{***}	0.0017^{***}	0.0037	0.0004
	(0.0038)	(0.0032)	(0.0048)	(0.0005)	(0.0004)	(0.0005)	(0.0032)	(0.0004)
Individual treated × neighbors' reform exposure			−0.0039			−0.0006		
			(0.0058)			(0.0007)		
Scaled Spillover Effects:								
Average		0.0579^{**}			0.1265^{***}			
		(0.0273)			(0.0334)			
Individual untreated			0.0777^{*}			0.1581^{***}		
			(0.0416)			(0.0477)		
Individual treated			0.0439			0.1054		
			(0.0649)			(0.0781)		
Observations	1,314,468	1,314,468	1,314,468	1,314,468	1,314,468	1,314,468	1,309,875	1,309,875
R-squared	0.265	0.059	0.059	0.232	0.040	0.040	0.059	0.040
F-statistic (first stage)	936.29			507.73				

Notes: The table shows results from the instrumental variable estimation for neighbors. The upper panel displays regression results, with the respective outcomes indicated by the column titles. All specifications include fixed effects for wider neighborhoods (4-digit postcodes) and gender interacted with month-by-year of birth fixed effects as controls. The lower panel shows implied spillover effects scaled relative to the respective first-stage effect. Standard errors clustered by 5-digit postcode areas in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE 4: INSTRUMENTAL VARIABLE ESTIMATES FOR COWORKERS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Main Results			Social Norms			Placebo Results	
	First Stage: Coworkers' retirement age	*Spillover:* Individual retirement age		*First Stage:* Fraction of coworkers retiring at NRA	*Spillover:* Probability to retire at NRA		*Spillover:* Individual retirement age	*Spillover:* Probability to retire at NRA
Coworkers' reform exposure (standardized)	0.2122***	0.0222***	0.0622***	0.0217***	0.0022**	0.0085***	0.0044	0.0003
	(0.0060)	(0.0061)	(0.0085)	(0.0010)	(0.0011)	(0.0014)	(0.0054)	(0.0009)
Individual treated × coworkers' reform exposure			−0.0832***			−0.0131***		
			(0.0113)			(0.0018)		
Scaled Spillover Effects:								
Average		0.1045***			0.1009**			
		(0.0290)			(0.0502)			
Individual untreated			0.2465***			0.3019***		
			(0.0345)			(0.0502)		
Individual treated			−0.0835			−0.1637*		
			(0.0576)			(0.0846)		
Observations	594,613	594,613	594,613	594,613	594,613	594,613	537,520	537,520
R-squared	0.178	0.078	0.078	0.212	0.056	0.056	0.079	0.059
F-statistic (first stage)	1264.18			468.68				

Notes: The table shows results from the instrumental variable estimation for coworkers. The upper panel displays regression results, with the respective outcomes indicated by the column titles. All specifications include sector fixed effects and gender interacted with month-by-year of birth fixed effects as controls. The lower panel shows implied spillover effects scaled relative to the respective first-stage effect. Standard errors clustered by firm in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

TABLE 5: THE SOCIAL MULTIPLIER OF PENSION REFORM

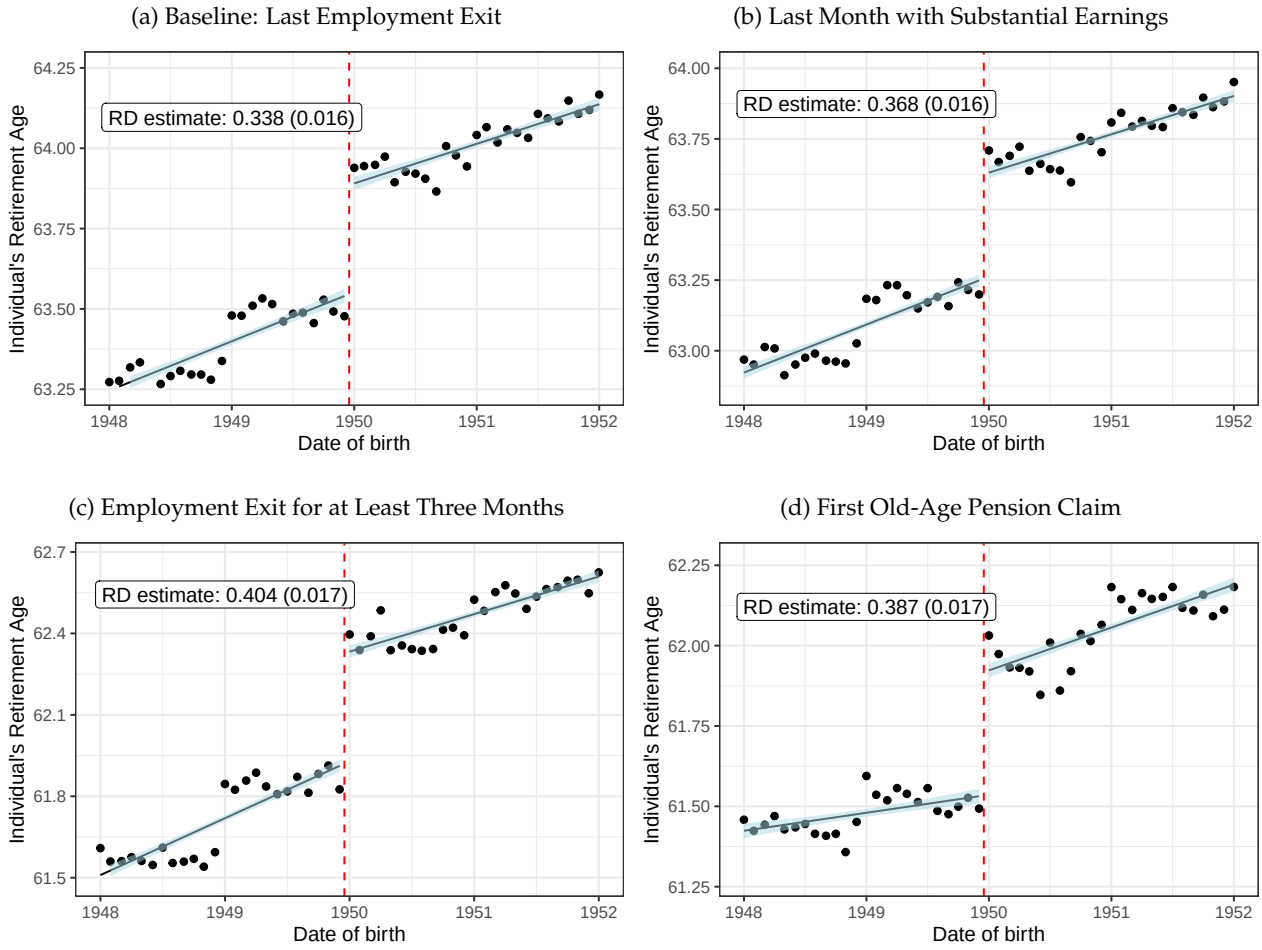
	(1)	(2)	(3)	(4)	(5)	(6)
	Direct	Group-specific multiplier				Total
	effect	Spouses	Siblings	Neighbors	Coworkers	multiplier
Multiplier on:						
Average retirement age	1.000	0.189	0.070	0.058	0.104	1.422
		(0.034)	(0.023)	(0.027)	(0.029)	(0.057)
Total earnings	1.000	0.113	0.071	0.059	0.105	1.347
		(0.048)	(0.027)	(0.028)	(0.029)	(0.067)
Total fiscal impact	1.000	0.107	0.071	0.059	0.105	1.342
		(0.052)	(0.027)	(0.028)	(0.029)	(0.071)

Notes: The table shows the social multiplier of pension reform on each of the outcomes indicated by the row titles. As the first column indicates, we consider a pension reform with a direct effect normalized to one. Columns (2) to (5) display group-specific multipliers for spouses, siblings, neighbors, and coworkers, quantifying each group's contribution to the total multiplier. The last column shows the total social multiplier aggregated across all peer groups. Standard errors in parentheses are computed using the delta method.

Online Appendix

A Appendix Figures and Tables

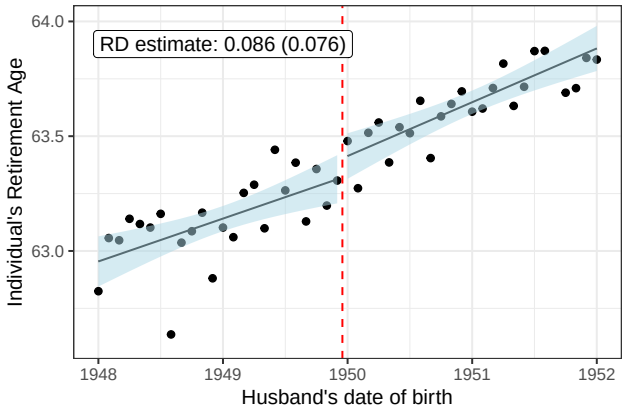
FIGURE A1: ALTERNATIVE RETIREMENT DEFINITIONS



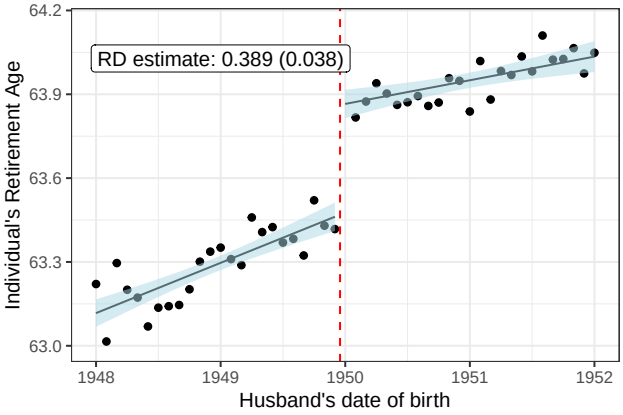
Notes: The figure shows retirement patterns and the estimated impact of the 2006 reform under alternative retirement definitions. Panel (a) shows our baseline specification, where retirement is defined as an individual's last employment exit. Panel (b) defines retirement as the last month with substantial earnings ($\geq$ EUR 500), thus allowing for some earnings after retirement. Panel (c) considers retirement as employment exits after which the individual does not work for at least three consecutive months. Panel (d) defines retirement as the first month of claiming an old-age pension. Each panel shows individuals' average retirement ages, according to the respective definition, by their own birth date in monthly bins. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

FIGURE A2: SPILLOVERS IN COUPLES

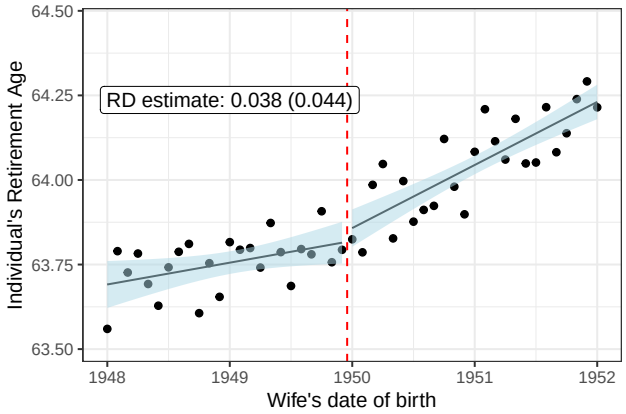
(a) Older Women



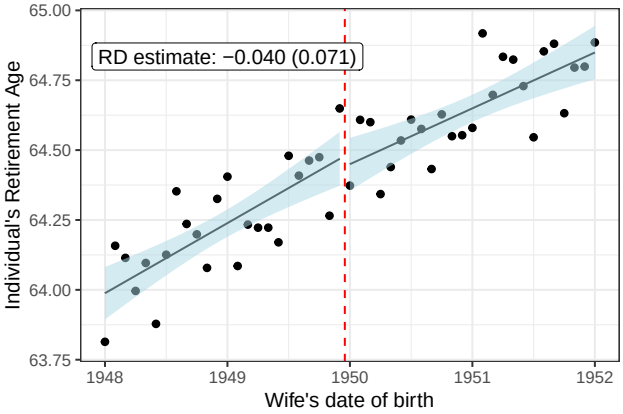
(b) Younger Women



(c) Older Men

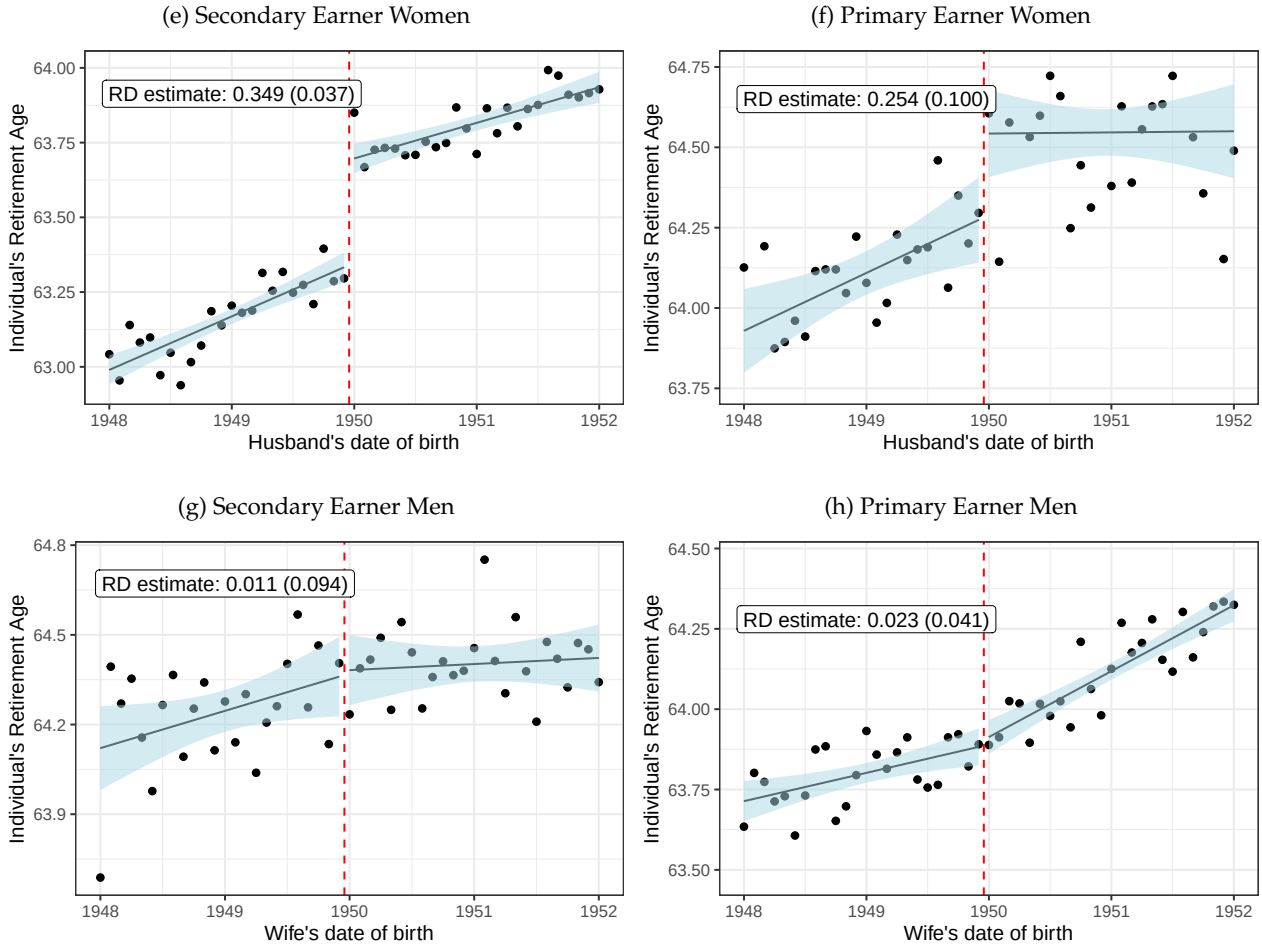


(d) Younger Men



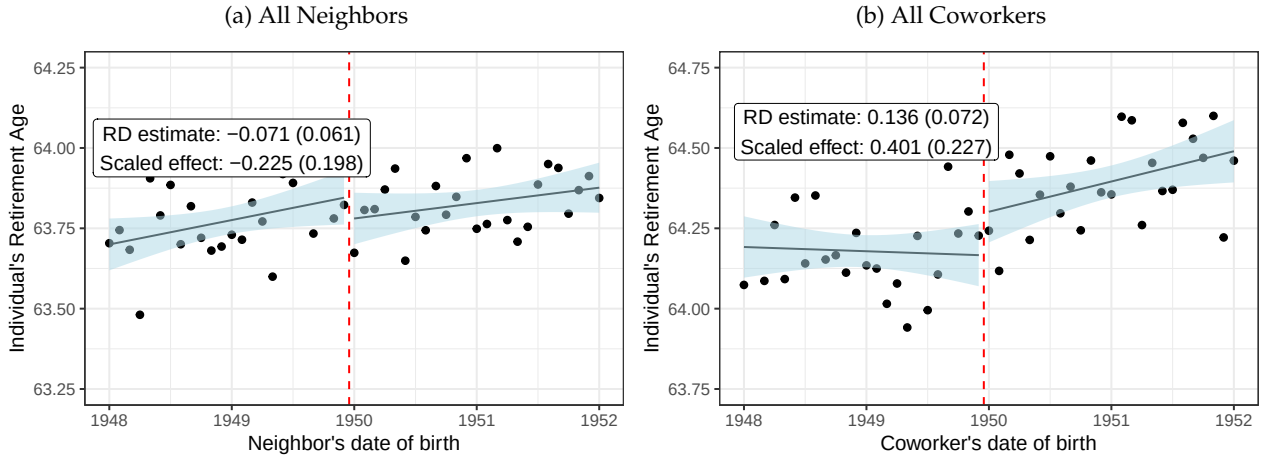
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FIGURE A2: SPILLOVERS IN COUPLES (CONTINUED)



Notes: The figure shows individuals' average retirement ages by their spouse's birth date in monthly bins. The respective sample is indicated by the panel titles. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

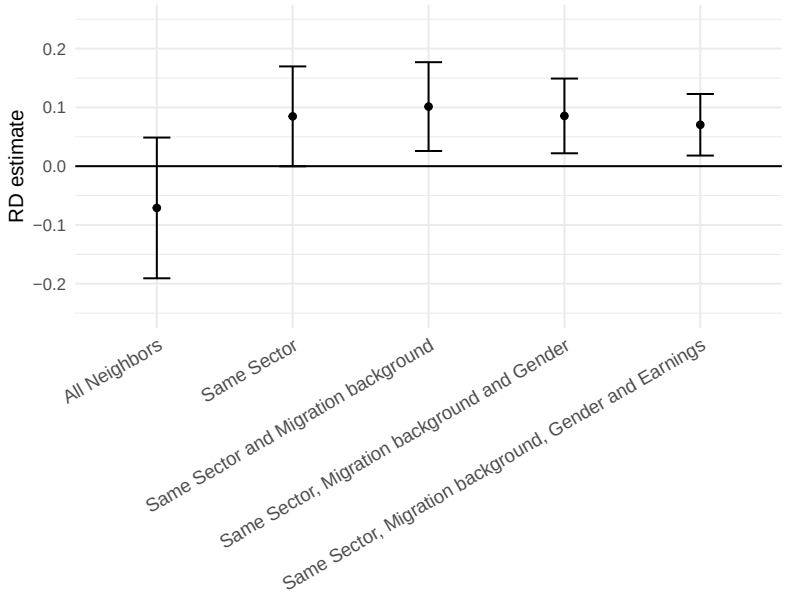
FIGURE A3: RD ESTIMATES OF AVERAGE SPILLOVERS BETWEEN NEIGHBORS AND COWORKERS



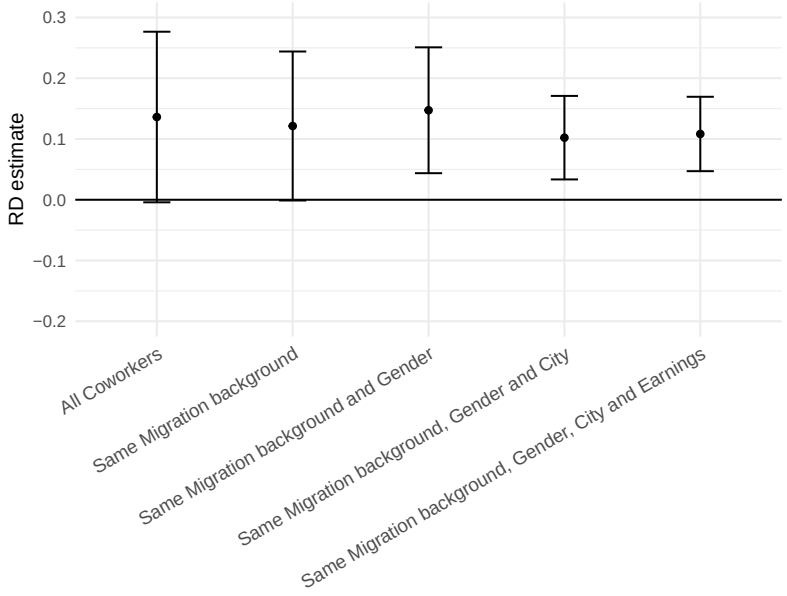
Notes: The figure shows individuals' average retirement ages by their neighbor's or coworker's birth date in monthly bins. In Panel (a), neighborhood peers are selected out of all neighbors, and in Panel (b), workplace peers are selected out of all coworkers. In each panel, the red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. The text boxes display the reduced-form RD estimate of the jump in retirement ages at the cutoff as well as the implied scaled spillover effect, with standard errors in parentheses. An extended set of RD estimates for neighbors and coworkers are shown in Appendix Figure A4.

FIGURE A4: EXTENDED RESULTS FOR NEIGHBORS AND COWORKERS

(a) Neighbors

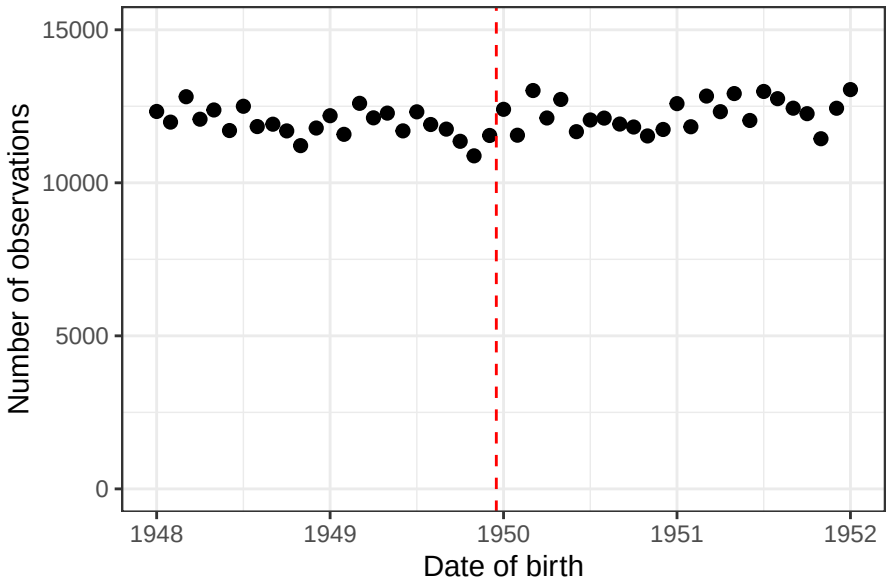


(b) Coworkers



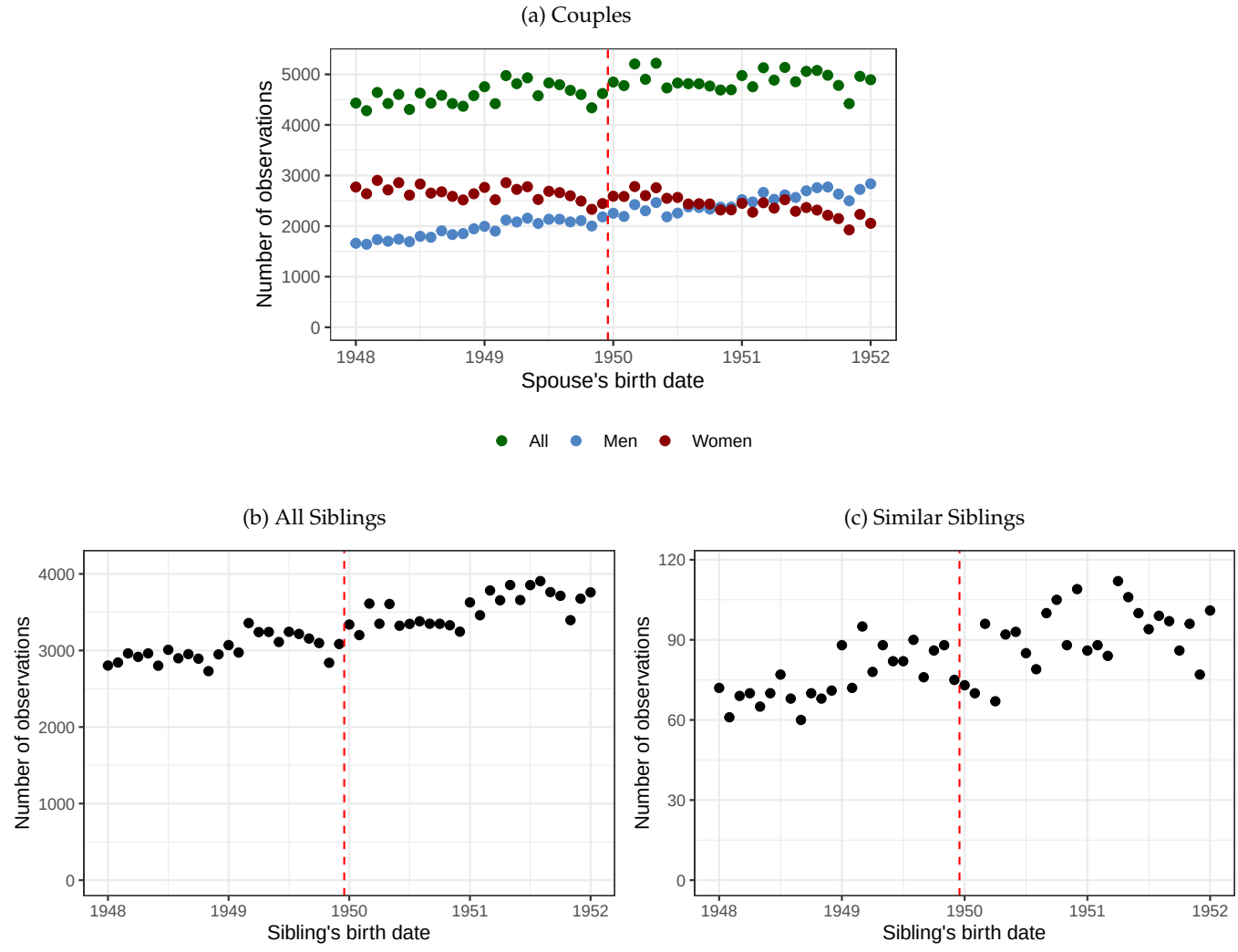
Notes: The figure shows an extended set of regression discontinuity estimates for neighbors and coworkers. Both panels display spillover effects of a peer's reform exposure on an individual's retirement age estimated via a regression discontinuity design, with 95% confidence intervals. The respective sub-samples of neighbors or coworkers are indicated by the labels on the horizontal axis.

FIGURE A5: DISTRIBUTION OF BIRTH DATES (FULL SAMPLE)



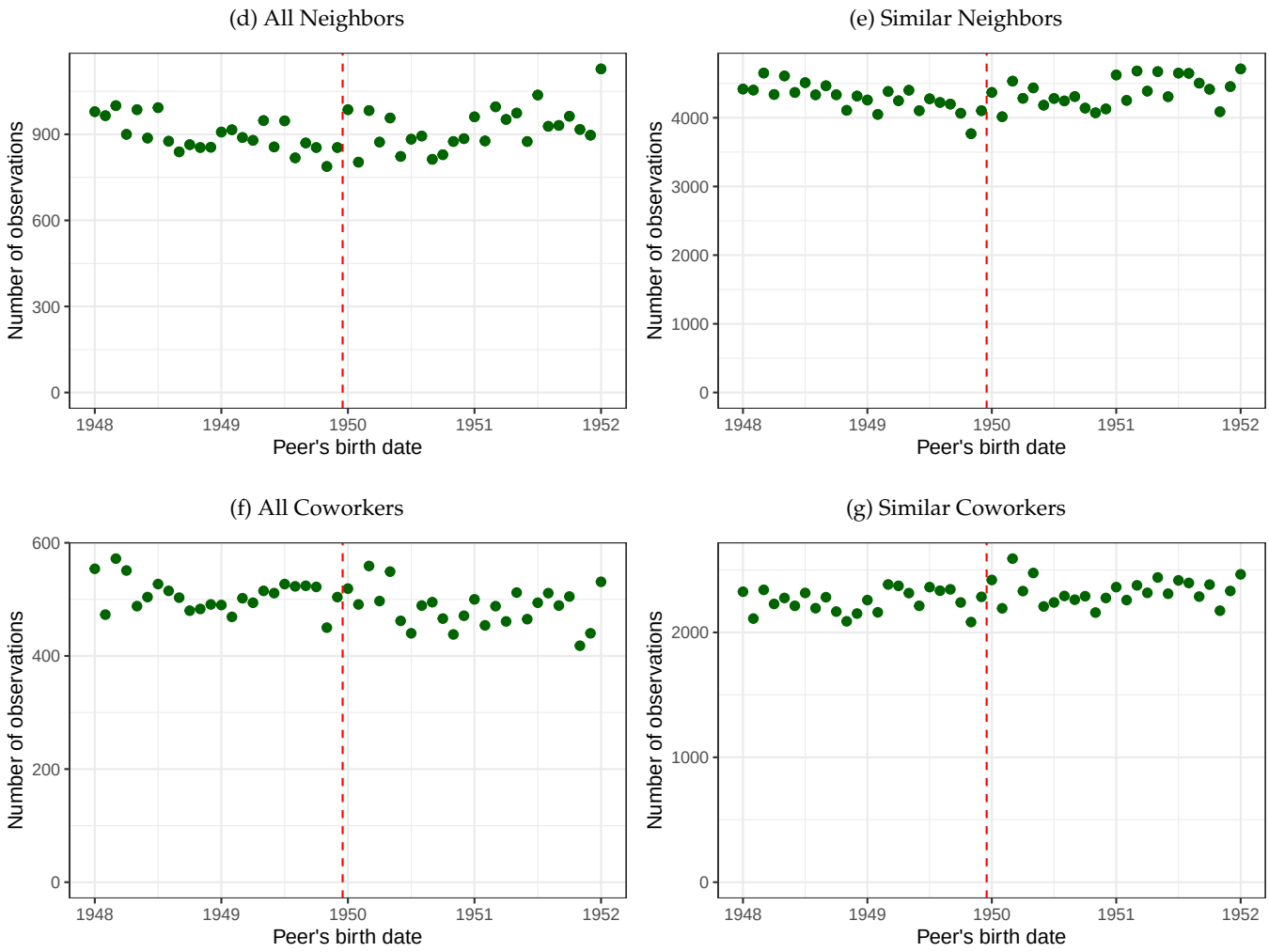
Notes: The figure shows the distribution of birth dates, the running variable for our regression discontinuity design, in monthly bins within the full sample. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform.

FIGURE A6: DISTRIBUTION OF BIRTH DATES IN THE RD SAMPLES



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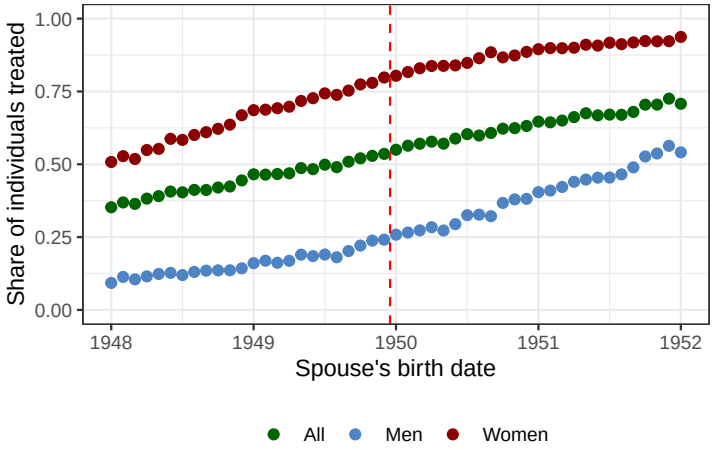
FIGURE A6: DISTRIBUTION OF BIRTH DATES IN THE RD SAMPLES (CONTINUED)



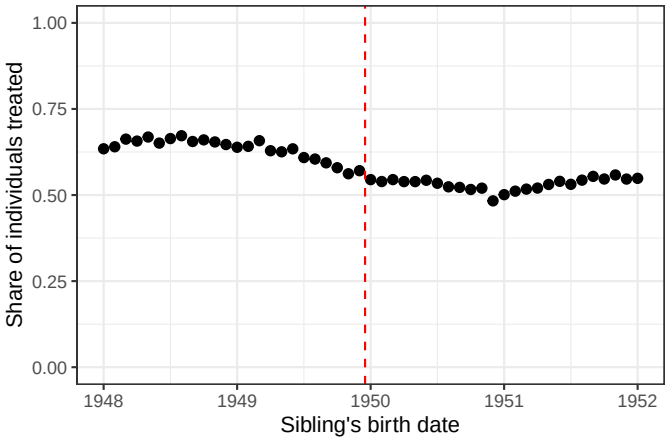
Notes: The figure shows the distribution of birth dates, the running variable for our regression discontinuity design, in monthly bins for each of the samples underlying our main regression discontinuity estimates. Panel titles and figure legends indicate the respective sample. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform.

FIGURE A7: REFORM EXPOSURE CORRELATION IN THE RD SAMPLES

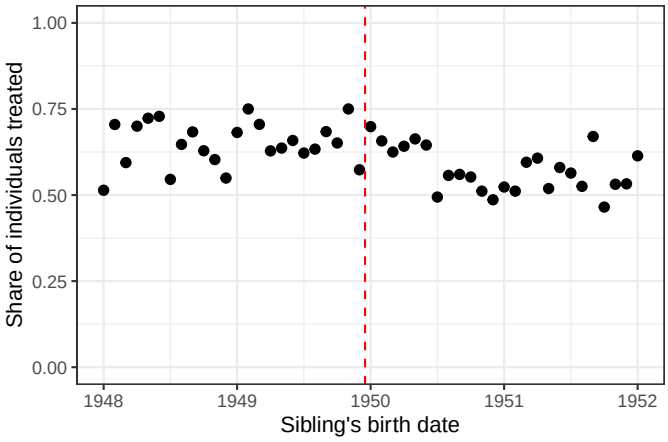
(a) Couples



(b) All Siblings

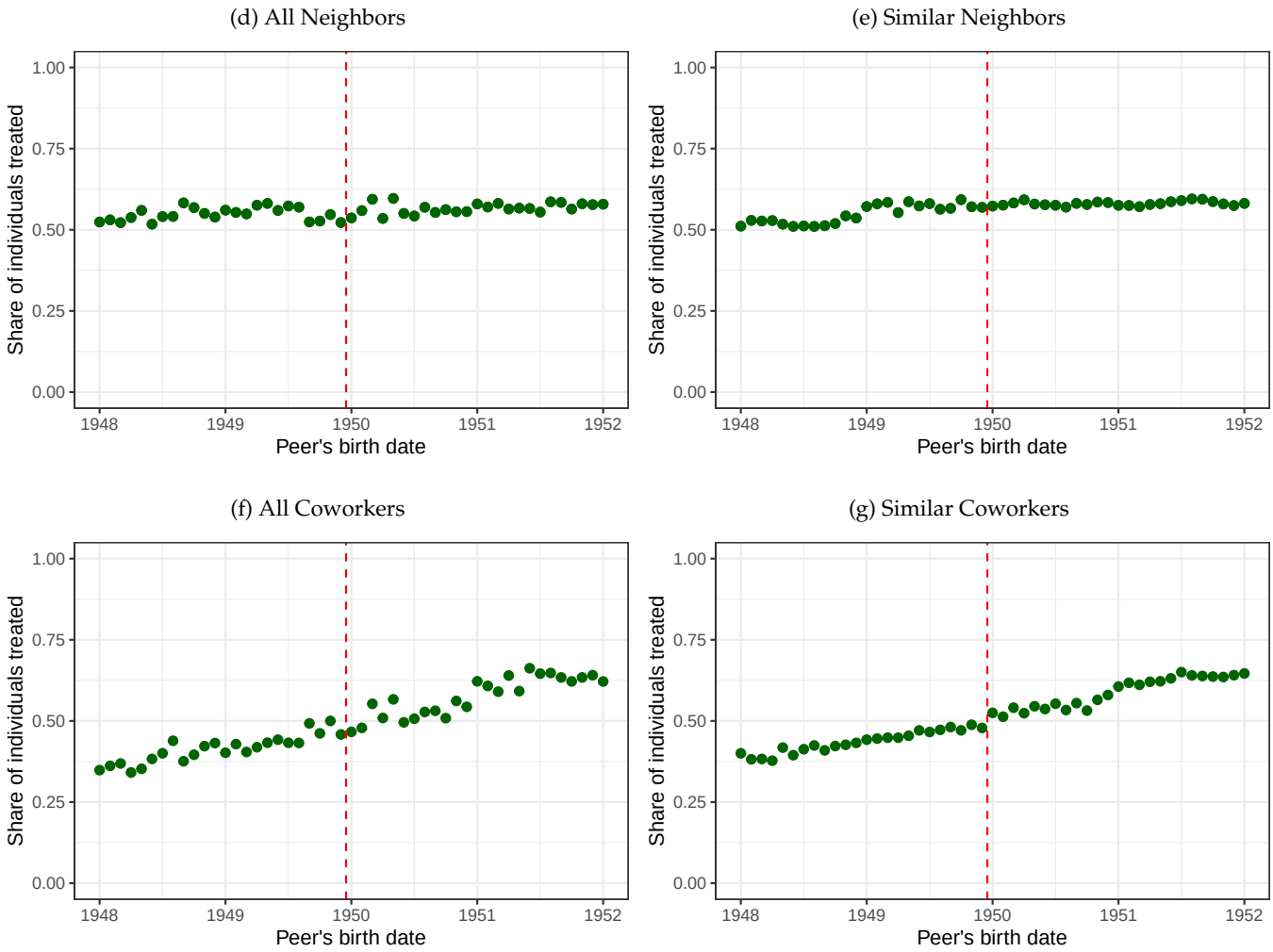


(c) Similar Siblings



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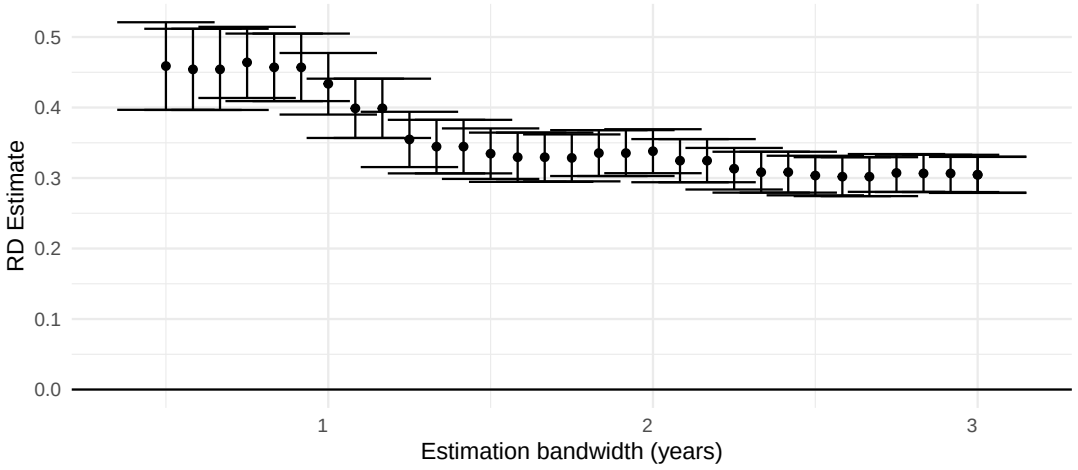
FIGURE A7: REFORM EXPOSURE CORRELATION IN THE RD SAMPLES (CONTINUED)



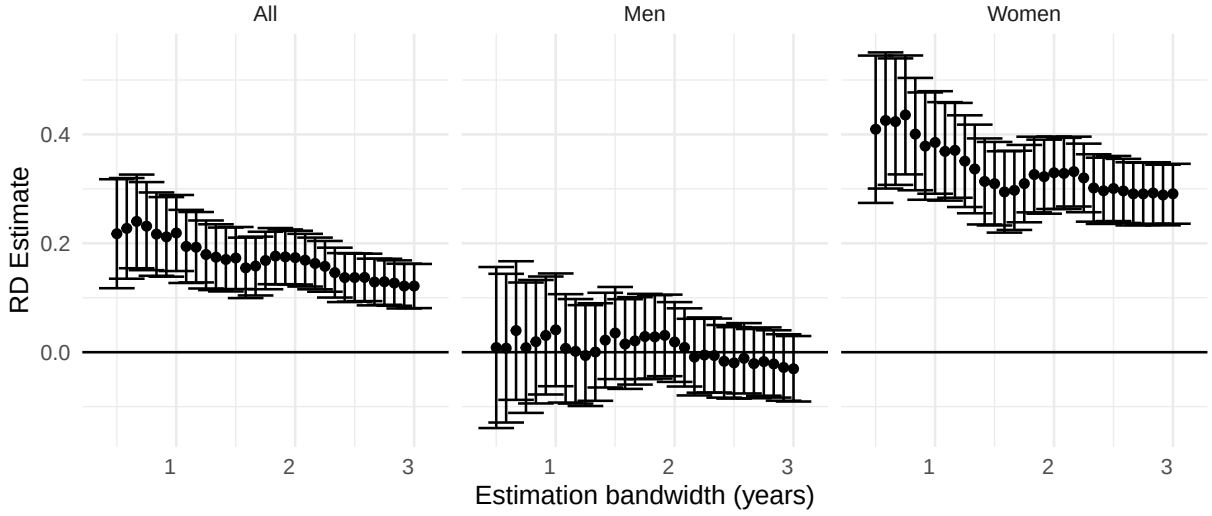
Notes: The figure shows the correlation of individuals' reform exposure and peers' birth date for each of the samples underlying our main regression discontinuity estimates. Each panel displays the share of individuals exposed to the reform, i.e. born above the January 1950 cutoff, by the peer's birth date in monthly bins. Panel titles and figure legends indicate the respective sample. The red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. See Column (8) of Appendix Table A2 for corresponding balancing tests of individual reform exposure around the cutoff.

FIGURE A8: REGRESSION DISCONTINUITY ESTIMATES BY BANDWIDTH

(a) First Stage (Full Sample)



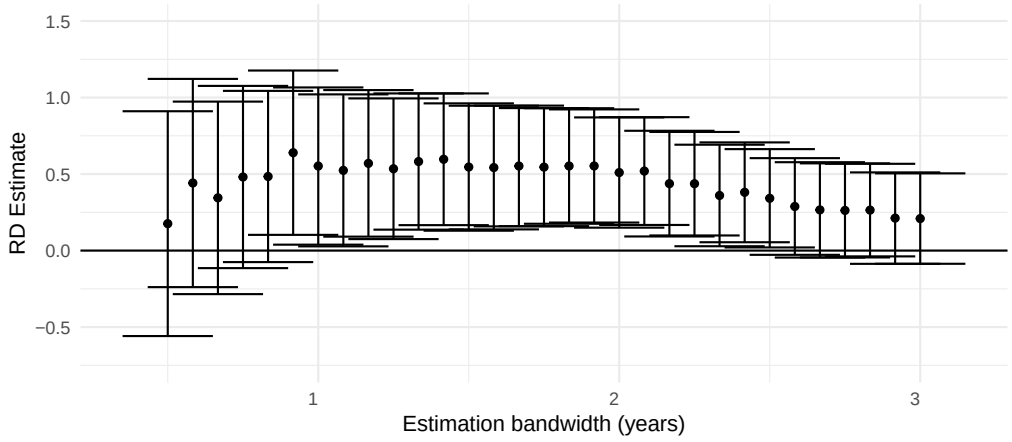
(b) Couples



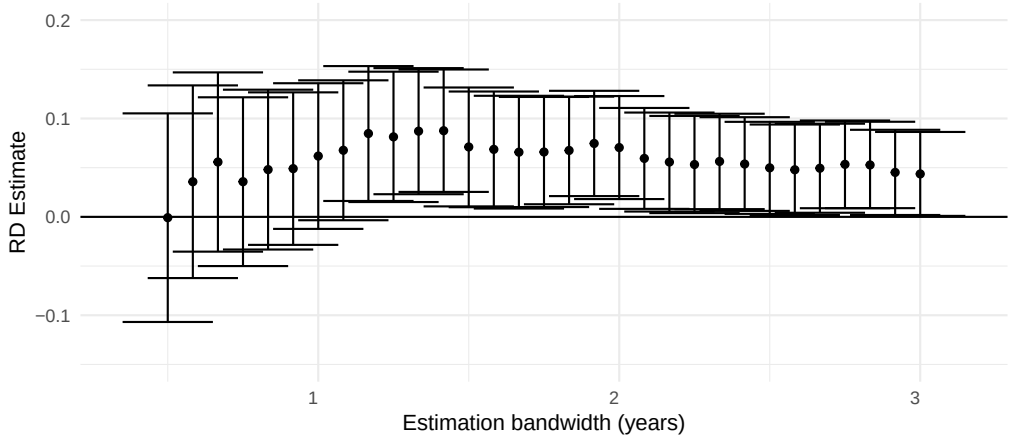
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FIGURE A8: REGRESSION DISCONTINUITY ESTIMATES BY BANDWIDTH (CONTINUED)

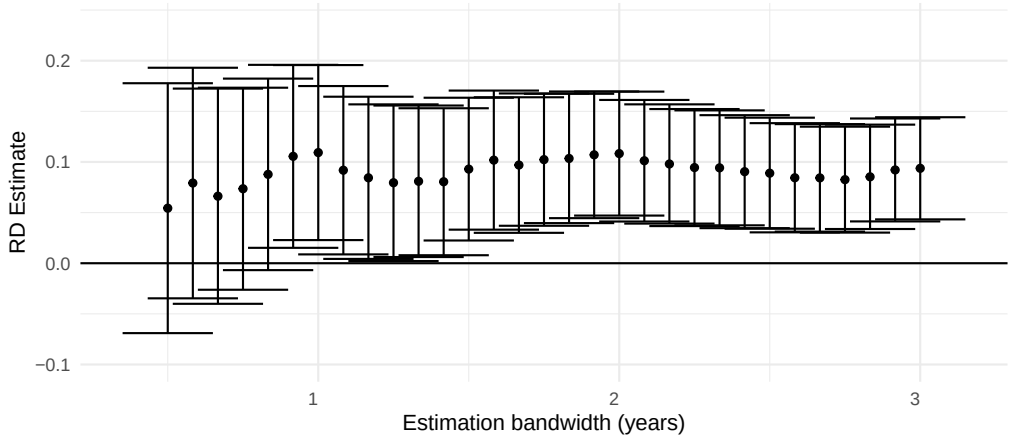
(c) Similar Siblings



(d) Similar Neighbors

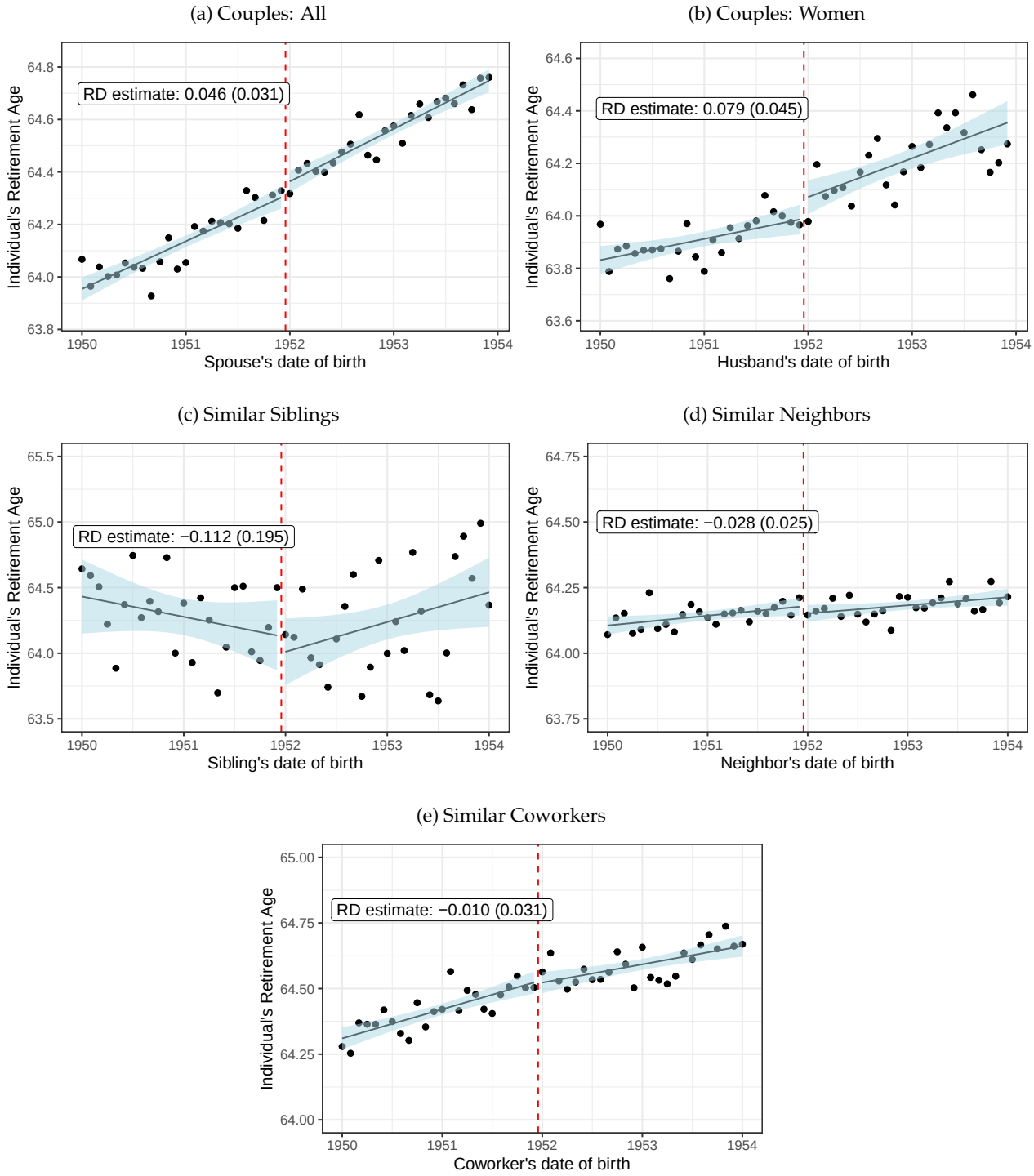


(e) Similar Coworkers



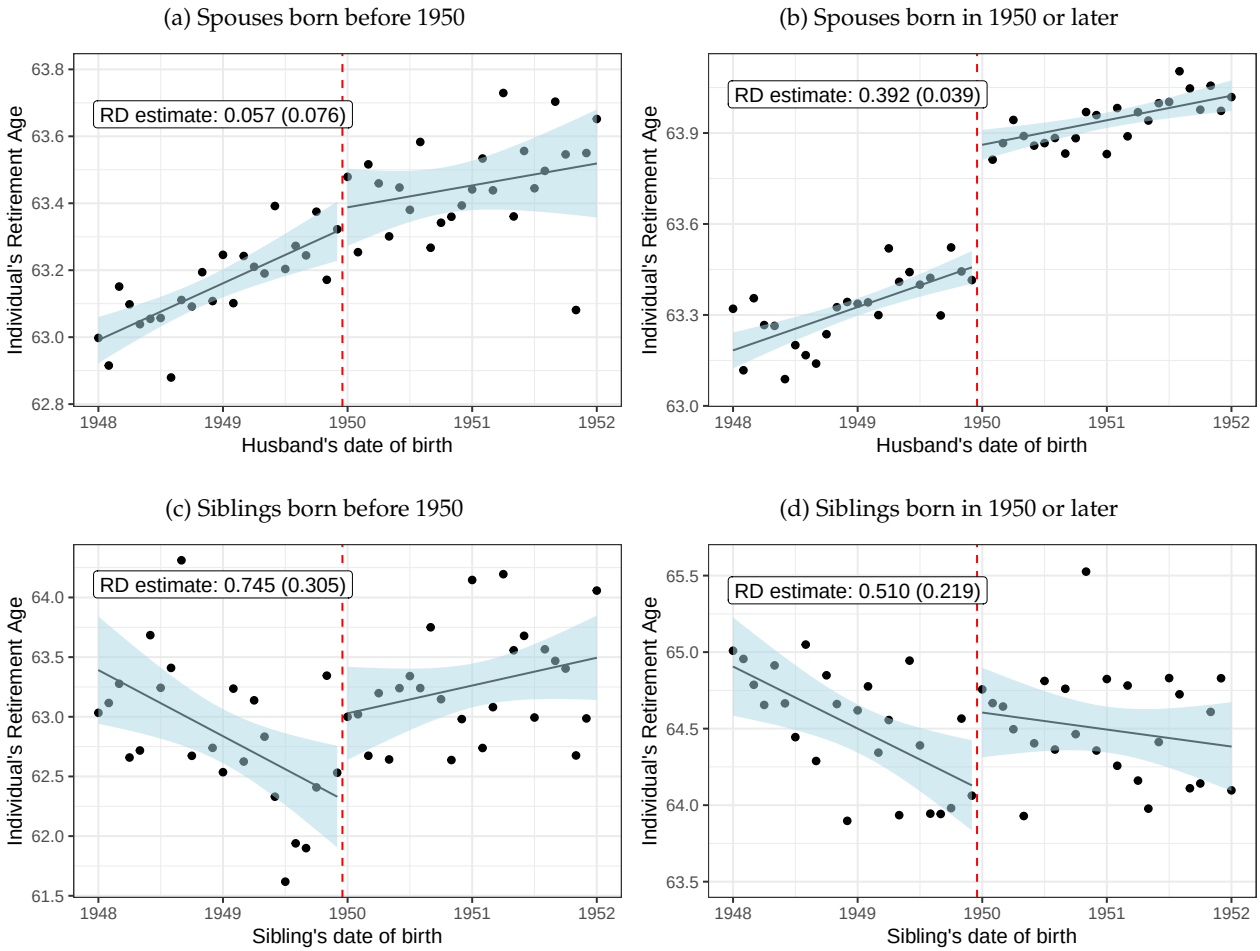
Notes: The figure shows regression discontinuity estimates for a range of estimation bandwidths between 0.5 and 3 years in monthly increments. Panel titles and figure legends indicate the respective sample and peer group. Other than varying the bandwidth, all estimation parameters are the same as in the main specifications from Table 2.

FIGURE A9: REGRESSION DISCONTINUITY PLACEBO CHECKS



Notes: The figure shows placebo regression discontinuity design specifications around an artificial birth date cutoff in January 1952. Each panel displays individuals' average retirement ages by their peer's birth date in monthly bins. The respective placebo cutoff and sample are indicated by the panel titles. The red dashed line demarcates the placebo birth date cutoff. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

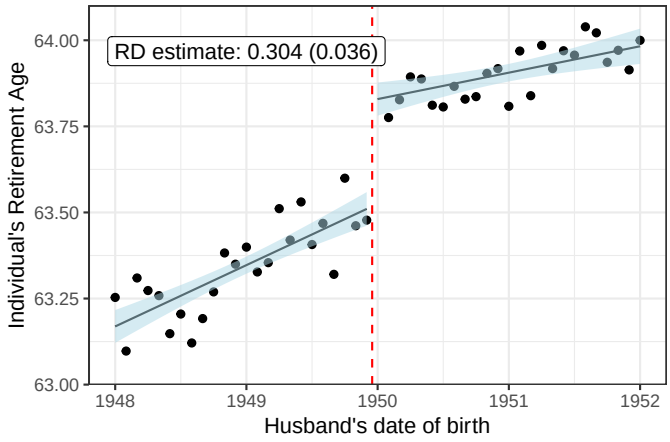
FIGURE A10: SPILLOVER EFFECTS BY OWN REFORM EXPOSURE



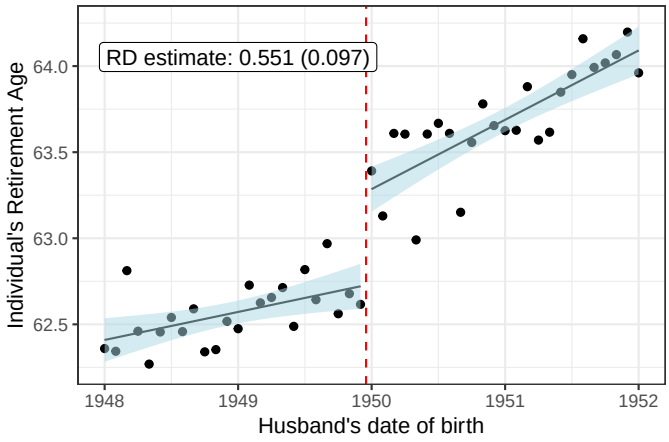
Notes: The figure shows individuals' average retirement ages by their spouse's or sibling's birth date in monthly bins. The respective sample is split by individuals' own reform exposure, as indicated by the panel titles. In each panel, the red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

FIGURE A11: THE ROLE OF JOINT RETIREMENT

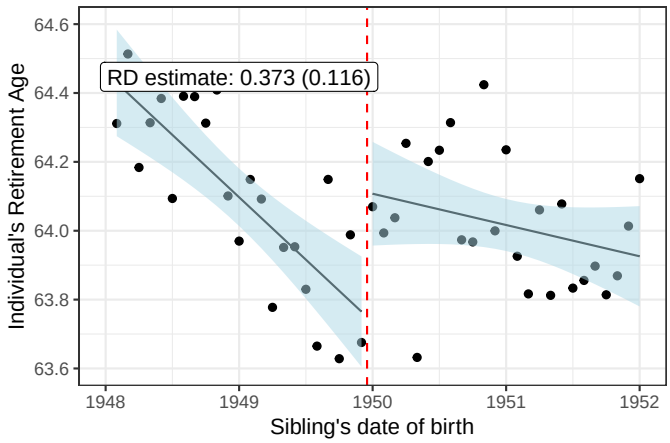
(a) Couples retiring separately



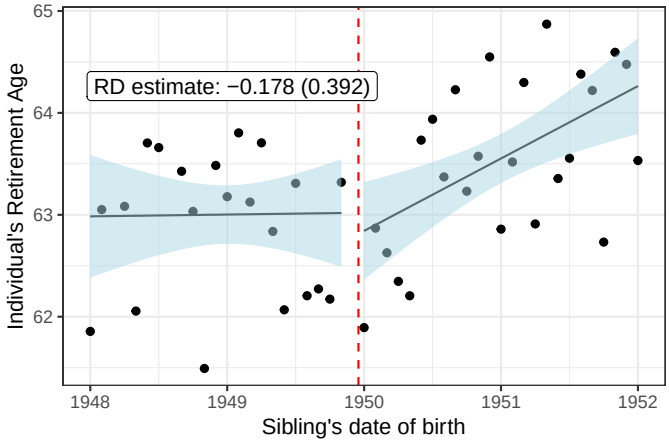
(b) Couples retiring jointly



(c) Siblings retiring separately

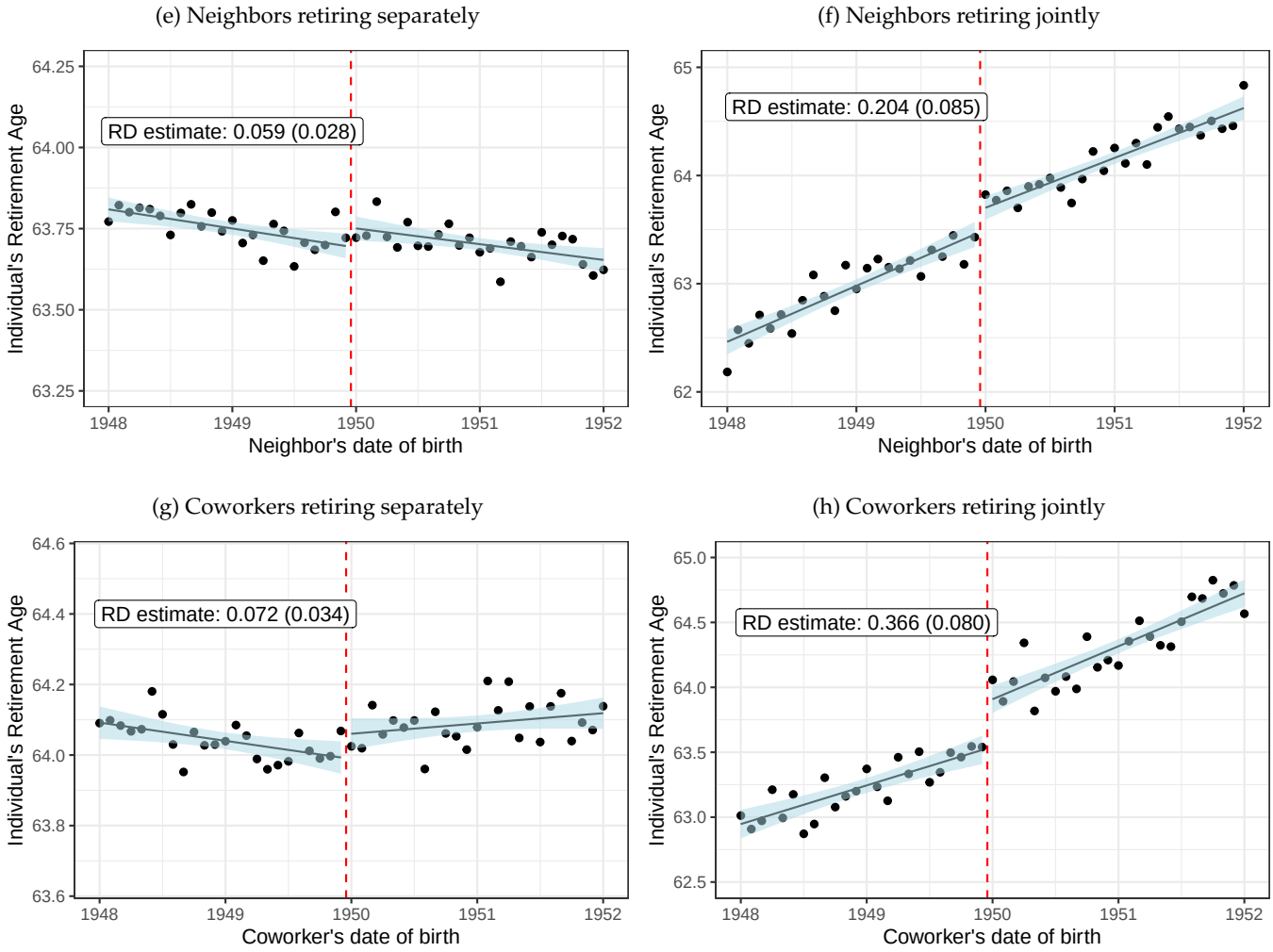


(d) Siblings retiring jointly



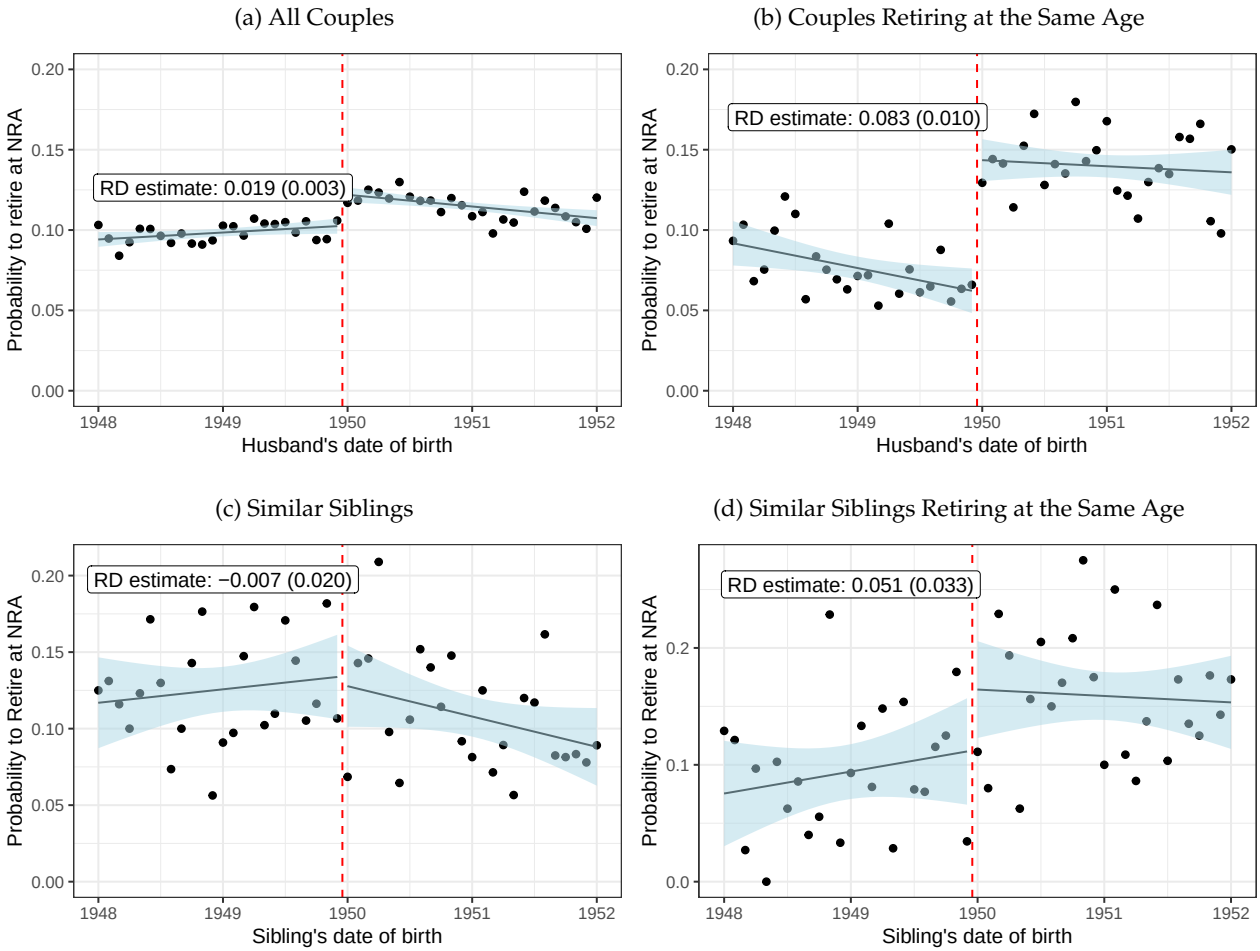
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FIGURE A11: THE ROLE OF JOINT RETIREMENT (CONTINUED)



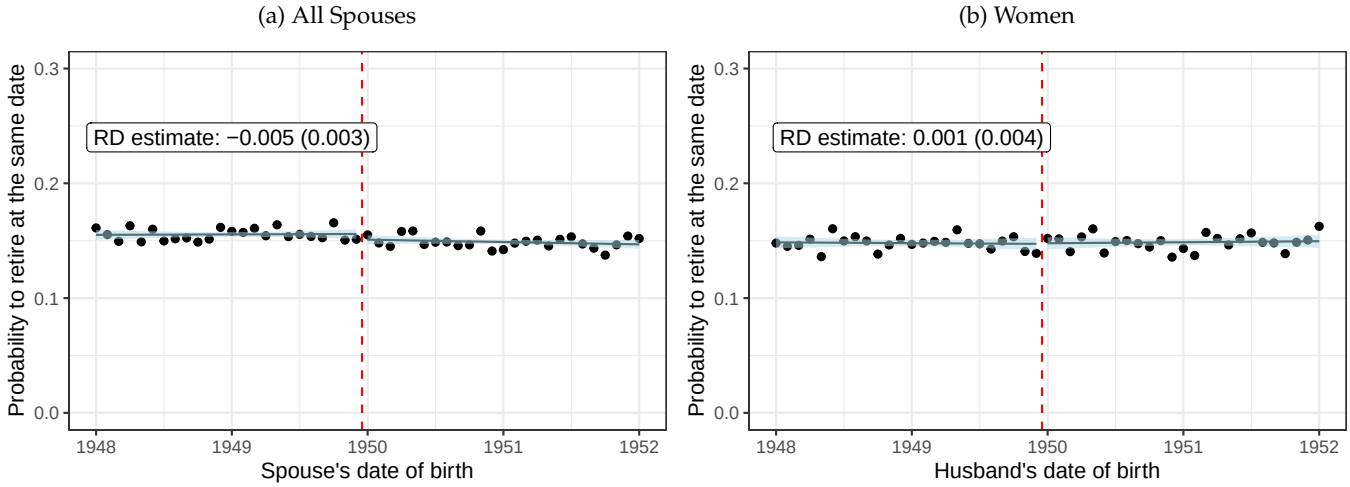
Notes: The figure shows individuals' average retirement ages by their peer's birth date in monthly bins. The respective sample is split into pairs retiring jointly vs. pairs retiring separately, as indicated by the panel titles. Joint retirement is defined as individuals retiring within +/- 6 months of the peer's retirement date, and separate retirement is defined as individuals retiring outside of this window. In each panel, the red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

FIGURE A12: PROPAGATION OF SOCIAL NORMS



Notes: The figure shows individuals' probability of retiring at the NRA by their spouse's or sibling's birth date in monthly bins. The respective sample is indicated by the panel titles. Retiring at the same age is defined as retiring within +/- 3 months of the peer's retirement age. In each panel, the red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

FIGURE A13: COUPLES' JOINT RETIREMENT PROPENSITY AROUND THE REFORM CUTOFF



Notes: The figure shows individuals' probability to retire jointly with their spouse by the spouse's birth date in monthly bins. The respective sample is indicated by the panel titles. Joint retirement is defined as individuals retiring within +/- 6 months of the peer's retirement date. In each panel, the red dashed line demarcates the January 1950 cutoff of the 2006 pension reform. The black solid lines depict a linear fit estimated separately on each side of the cutoff, with 95% confidence bands shaded in light blue. An estimate of the jump in retirement ages at the cutoff from a regression discontinuity design is included in the text box, with its standard error in parentheses.

TABLE A1: HETEROGENEITY IN SPILLOVERS BETWEEN NEIGHBORS AND COWORKERS

	(1)	(2)		(3)	(4)
	Neighbors (by Population Density)			Coworkers (by Firm Size)	
	First Stage: Neighbors' retirement age	*Spillover:* Individual retirement age		*First Stage:* Coworkers' retirement age	*Spillover:* Individual retirement age
Neighbors' reform exposure (standardized)	0.1002^{***}	0.0169^{**}	Coworkers' reform exposure (standardized)	0.1539^{***}	0.0325^{***}
	(0.0086)	(0.0072)		(0.0082)	(0.0082)
× density category 2	0.0222^{**}	-0.0045	× Firm-size quintile 2	0.0611^{***}	-0.0027
	(0.0112)	(0.0096)		(0.0109)	(0.0111)
× density category 3	0.0165	-0.0195^{*}	× Firm-size quintile 3	0.0955^{***}	-0.0074
	(0.0126)	(0.0107)		(0.0145)	(0.0148)
× density category 4	0.0255^{**}	-0.0134	× Firm-size quintile 4	0.1051^{***}	0.0001
	(0.0127)	(0.0109)		(0.0182)	(0.0186)
× density category 5 (least dense)	0.0174	-0.0132	× Firm-size quintile 5 (largest)	0.0878^{***}	-0.0239
	(0.0110)	(0.0093)		(0.0310)	(0.0318)
Scaled Spillover Effects:			*Scaled Spillover Effects:*		
Density category 1 (most dense)		0.1687^{**}	Quintile 1 (smallest)		0.2115^{***}
		(0.0732)			(0.0544)
Density category 2		0.1018^{*}	Quintile 2		0.1390^{***}
		(0.0532)			(0.0356)
Density category 3		-0.0220	Quintile 3		0.1008^{**}
		(0.0676)			(0.0500)
Density category 4		0.0280	Quintile 4		0.1260^{*}
		(0.0655)			(0.0653)
Density category 5 (least dense)		0.0314	Quintile 5 (largest)		0.0359
		(0.0502)			(0.1275)
Observations	$1,299,274$	$1,299,274$	Observations	$579,758$	$579,758$
R-squared	0.2665	0.0591	R-squared	0.1870	0.0803

Notes: The table shows results from the instrumental variable estimation for neighbors and coworkers. Columns (1) and (2) display heterogeneous spillovers between neighbors by local population density, using the following density categories: 1 is extremely urbanized (more than 2500 addresses per square kilometer), 2 is strongly urbanized (1501 to 2500 addresses), 3 is moderately urbanized (1001 to 1500 addresses), 4 is hardly urbanized (501 to 1000 addresses), 5 is not urbanized (500 or fewer addresses). These specifications include fixed effects for wider neighborhoods (4-digit postcodes) and gender interacted with month-by-year of birth fixed effects as controls. Columns (3) and (4) display heterogeneous spillovers between coworkers by firm size quintiles, based on the total number of workers in a firm. These specifications include sector fixed effects and gender interacted with month-by-year of birth fixed effects as controls. The lower panel shows implied spillover effects scaled relative to the respective first-stage effect. Standard errors clustered by 5-digit postcode areas (Columns 1 and 2) or by firm (Columns 3 and 4) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE A2: REGRESSION DISCONTINUITY DESIGN – BALANCING TESTS

	(1) Dutch-born	(2) Married	(3) Female	(4) Earnings at 57	(5) No. of siblings	(6) No. of coworkers	(7) No. of neighbors	(8) Treated	(9) NRA
Panel A: First Stage, Full Sample (N=593,031)									
Own birth date above 1950 cutoff	-0.001	-0.001	-0.004	560.25**	-0.021***	11.565	-0.131		
	(0.002)	(0.002)	(0.003)	(240.350)	(0.007)	(8.158)	(0.175)		
Panel B: Couples Sample, All (N=232,244)									
Peer's birth date above 1950 cutoff	0.001		0.018***	-417.99*	-0.016	-1.018	-0.118	0.013***	0.019***
	(0.002)		(0.004)	(245.178)	(0.011)	(12.432)	(0.282)	(0.004)	(0.007)
Panel C: Couples Sample, Females (N=123,810)									
Peer's birth date above 1950 cutoff	0.004			247.74	-0.023	-11.946	0.050	0.007	0.022**
	(0.003)			(211.582)	(0.015)	(14.445)	(0.385)	(0.005)	(0.010)
Panel D: Siblings Sample, All (N=160,889)									
Peer's birth date above 1950 cutoff	0.003	0.002	0.002	294.29	-0.005*	2.807	-0.052	-0.063***	-0.011
	(0.003)	(0.004)	(0.005)	(382.561)	(0.003)	(15.858)	(0.332)	(0.005)	(0.010)
Panel E: Siblings Sample, Most Similar (N=4,104)									
Peer's birth date above 1950 cutoff	-0.011	0.026	-0.007	4513.65***	-0.088	-0.093	-2.656	-0.052*	0.002
	(0.016)	(0.025)	(0.029)	(1569.120)	(0.074)	(93.373)	(2.126)	(0.031)	(0.046)
Panel F: Neighbors Sample, All (N=44,565)									
Peer's birth date above 1950 cutoff	-0.002	0.015	-0.002	509.96	-0.017	-15.988	-0.099	-0.001	0.026
	(0.007)	(0.009)	(0.009)	(659.406)	(0.024)	(30.180)	(0.455)	(0.010)	(0.020)
Panel G: Neighbors Sample, Similar (N=212,323)									
Peer's birth date above 1950 cutoff	-0.000	-0.005	0.001	187.01	-0.002	1.510	-0.451*	-0.013***	-0.017*
	(0.003)	(0.004)	(0.004)	(325.922)	(0.011)	(12.846)	(0.273)	(0.004)	0.009
Panel H: Coworkers Sample, All (N=24,316)									
Peer's birth date above 1950 cutoff	-0.018**	-0.024**	0.011	1660.27	-0.038	-1.399	-0.715	0.012	-0.001
	(0.008)	(0.010)	(0.012)	(1050.610)	(0.034)	(2.958)	(0.833)	(0.013)	(0.013)
Panel I: Coworkers Sample, Similar (N=112,307)									
Peer's birth date above 1950 cutoff	-0.005	0.000	-0.000	902.53*	-0.006	4.182	0.246	0.022***	-0.003
	(0.004)	(0.005)	(0.006)	(532.312)	(0.016)	(10.789)	(0.392)	(0.006)	0.006

Notes: The table shows balancing tests in the context of our regression discontinuity design, testing for discontinuities in predetermined covariates. Each coefficient corresponds to a regression discontinuity estimate for the outcome indicated by the respective column title. Panel A tests for discontinuities in individual characteristics around the January 1950 cutoff in terms of own birth date, akin to the first-stage specification from equation (1). Panels B to I test for discontinuities in individual characteristics around the January 1950 cutoff in terms of their peer's birth date, akin to equation (2) that we use to estimate spillover effects. This is done in each of our main RD samples, as indicated by the panel titles. In Column (9), "NRA" denotes the Normal Retirement Age faced by an individual. N denotes the number of observations in the estimation bandwidth. Standard errors are in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

TABLE A3: REGRESSION DISCONTINUITY DESIGN – SPECIFICATION CHECKS

	(1) Baseline	(2) Smaller bandwidth	(3) Larger bandwidth	(4) Quadratic fit	(5) Triangular kernel
			Panel A: First Stage, Full Sample		
Own birth date	0.338***	0.434***	0.305***	0.383***	0.358***
above 1950 cutoff	(0.016)	(0.022)	(0.013)	(0.024)	(0.017)
Bandwidth (years)	2	1	3	2	2
Observations	593,031	299,487	897,489	593,031	567,654
			Panel B: Spillovers in Couples, All		
Peer's birth date	0.174***	0.219***	0.122***	0.188***	0.180***
above 1950 cutoff	(0.025)	(0.036)	(0.021)	(0.038)	(0.028)
Bandwidth (years)	2	1	3	2	2
Observations	232,244	119,610	338,258	232,244	222,919
			Panel C: Spillovers in Couples, Women		
Peer's birth date	0.330***	0.385***	0.291***	0.340***	0.335***
above 1950 cutoff	(0.034)	(0.048)	(0.028)	(0.052)	(0.038)
Bandwidth (years)	2	1	3	2	2
Observations	123,810	64,238	179,205	123,810	118,983
			Panel D: Spillovers among Siblings, Similar		
Peer's birth date	0.510***	0.552**	0.209	0.601**	0.546***
above 1950 cutoff	(0.183)	(0.262)	(0.151)	(0.277)	(0.201)
Bandwidth (years)	2	1	3	2	2
Observations	4,104	2,143	6,038	4,104	3,931
			Panel E: Spillovers among Neighbors, Similar		
Peer's birth date	0.070***	0.062	0.044**	0.068*	0.070**
above 1950 cutoff	(0.027)	(0.038)	(0.022)	(0.040)	(0.029)
Bandwidth (years)	2	1	3	2	2
Observations	212,323	105,695	322,768	212,323	203,194
			Panel F: Spillovers among Coworkers, Similar		
Peer's birth date	0.108***	0.109**	0.094***	0.075	0.095***
above 1950 cutoff	(0.031)	(0.044)	(0.026)	(0.047)	(0.035)
Bandwidth (years)	2	1	3	2	2
Observations	112,307	57,456	166,458	112,307	107,516

Notes: The table shows regression discontinuity results under a range of specification choices. Panel A shows first-stage estimates using the full sample (like Figure 1), and Panels B to F show spillover effects (like Panel B of Table 2) for the respective groups indicated by the panel titles. Column (1) displays our baseline results, using linear fit, a uniform kernel, and a bandwidth of two years on each side of the January 1950 cutoff. Column (2) uses a smaller bandwidth of one year, Column (3) uses a larger bandwidth of three years, Column (4) uses a quadratic fit, and Column (5) uses a triangular kernel. "Observations" denotes the number of observations within the estimation bandwidth in each sample and specification. Standard errors are in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

TABLE A4: SOCIAL MULTIPLIER CALCULATION – ROBUSTNESS

	(1)	(2)	(3)	(4)	(5)	(6)
	Direct	Group-specific multiplier				Total
	effect	Spouses	Siblings	Neighbors	Coworkers	multiplier
		Multiplier on the average retirement age				
Baseline specification	1.000	0.189	0.070	0.058	0.104	1.422
		(0.034)	(0.023)	(0.027)	(0.029)	(0.057)
Broad peer sample	1.000	0.229	0.095	0.058	0.104	1.486
		(0.041)	(0.032)	(0.027)	(0.029)	(0.065)
Very broad peer sample	1.000	0.327	0.133	0.058	0.104	1.623
		(0.058)	(0.044)	(0.027)	(0.029)	(0.083)
Only RD estimates	1.000	0.189	0.070	0.029	0.057	1.346
		(0.034)	(0.023)	(0.011)	(0.017)	(0.046)
Only average effects	1.000	0.185	0.102	0.058	0.104	1.449
		(0.030)	(0.094)	(0.027)	(0.029)	(0.107)

Notes: The table shows the social multiplier of pension reform under a range of specification choices. The first row corresponds to the baseline specification from Table 5. The second row assumes that spillovers occur for a broader range of peer cohorts, namely 1942 to 1954. The third row assumes that spillovers occur for a very broad peer sample, additionally relaxing the restriction that peers are observed to work at or after age 57. The fourth row uses only regression discontinuity estimates for all groups instead of using IV estimates for neighbors and coworkers. The fifth row uses estimated average spillover effects for all groups instead of using subgroup effects for spouses and siblings. As the first column shows, we consider a pension reform with a direct effect on the average retirement age normalized to one. Columns (2) to (5) display group-specific multipliers for spouses, siblings, neighbors, and coworkers, illustrating each group's contribution to the total multiplier. The last column shows the total social multiplier aggregated across all peer groups. Standard errors in parentheses are computed using the delta method.

B Administrative Data

In this paper, we use administrativfe data covering the universe of Dutch residents provided by Statistics Netherlands (CBS). The data consists of a number of individual- and household-level registers, which are accessible via a remote access environment. Registers can be linked at the individual level via pseudonymized individual identifier numbers. We use civil status histories, which contain panel information on marriages, civil partnerships, separations, divorce, and widow(er)hood. We link individuals to their spouses and children based on a separate register dataset on family links. A dataset based on municipal registries contains information on individual addresses, which we can utilize to link individuals to their neighbors. We also use data on individual income sources, which are available from 1999 onward. This data provides monthly information on the type and amount of income. Finally, we use employer-employee matched labor market data, which is available from 2006 onward, in order to link individuals to their coworkers. This data also contains information on wages and hours worked.

In the following, we provide more information about the main datasets we use. A documentation of the respective dataset can be accessed by clicking on the links embedded in the names. Table B1 displays a full list of all data sources.[24]

GBAPERSOONTAB

This dataset contains demographic information, including gender, date of birth, and migration history, for the full Dutch population who have been registered in the Basic Register of Persons (BRP) since 1 October 1994.

GBAOVERLIJDENTAB

This dataset includes the date of death of all persons who were registered in the Basic Register of Persons (BRP) and died since 1 October 1994. It also includes the date of death of non-residents who were residents of the Netherlands at least once after 1 October 1994, and whose death information is available in the Register of Non-Residents (RNI). The dataset is primarily sourced from municipal registries (Gemeentelijke Basisadministratie Persoonsgegevens, GBA).

GBAADRESOBJECTBUS

For the universe of residents defined in the GBAPERSOONTAB data, this dataset contains information on individual addresses, with unique object identifiers at the level of buildings. The data is provided in a spell format, such that address changes can be observed. The main source of this data are municipal registries (Gemeentelijke Basisadministratie Persoonsgegevens, GBA).

[24] See the CBS website for a detailed catalogue of all available administrative datasets (in Dutch): https://www.cbs.nl/nl-nl/onze-diensten/maatwerk-en-microdata/microdata-zelf-onderzoek-doen/catalogus-microdata

POLISBUS and SPOLISBUS

These databases cover all employment spells in the Netherlands from 2006 onward. For each spell, they include information about the individual such as wage, hours worked, and social insurance contributions paid, employer identifiers, and additional information about the employer, such as sector of activity and collective bargaining agreements.

SECM datasets

These are a collection of datasets containing detailed information on individuals' monthly income by type, including wage earnings (SECMWERKNDGAMNBEDRABUS), profits from self-employment (SECMZLFMNDBEDRAGBUS), income from other activities (SECMOVACTMNDBEDRAGBUS), pension income (SECMPENSIOENMNDBEDRAGBUS), unemployment benefits (SECMWERKLMNDBEDRAGBUS), disability and sickness benefits (SECMZIEKTAOMNDBEDRAGBUS), welfare benefits (SECMBIJSTMNDBEDRAGBUS), and other benefits (SECMSOCVOORZOVMNDBEDRAGBUS). The information is available since 1999.

The SECM datasets are compiled by Statistic Netherlands drawing on various administrative data sources, including tax returns, social security data, and information from pension funds. The initial data is at the spell level, providing information about the start and end date of each income spell and the associated monthly amount of income. The SECMBUS data consolidates the different income sources into a single dataset containing the main source.

TABLE B1: ADMINISTRATIVE DATASETS

Key Variables Included	Name of Dataset	Years Used
Date of birth, gender	GBAPERSOON2023TAB (V1)	2023
Date of death	GBAOVERLIJDENTAB2023TAB (V1)	2023
Marital status	GBABURGERLIJKESTAAT2023BUS (V1)	2023
Household characteristics	GBAHUISHOUDENS2023BUS (V1)	2023
Residential location	GBAADRESOBJECT2023BUS (V1)	2023
Family links	KINDOUDER2023TAB (V1)	2023
Monthly income		
Wage earnings	SECMWERKNDGAMNBEDRABUSV20231	2023
Profits from self-employment	SECMZLFMNDBEDRAGBUSV20231	2023
Income from other activities	SECMOVACTMNDBEDRAGBUSV20231	2023
Pension income	SECMPENSIOENMNDBEDRAGBUSV20231	2023
Unemployment benefits	SECMWERKLMNDBEDRAGBUSV20231	2023
Disability and sickness benefits	SECMZIEKTAOMNDBEDRAGBUSV20231	2023
Welfare benefits	SECMBIJSTMNDBEDRAGBUSV20231	2023
Other benefits	SECMSOCVOORZOVMNDBEDRAGBUSV20231	2023
Firm, sector, wage, hours worked	POLISBUS and SPOLISBUS	2006–2023